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1.1 Introduction

1.1.1 Problem Statement

Greenfield University is currently running a multi-campus system that is not able to keep up with the needs of today's academic and administrative functions. The current network infrastructure has long-standing connectivity problems, security concerns, limited bandwidth, and is not scalable to meet the increasing number of users and new applications. Students and staff on all three campuses are frequently impacted by service interruptions, especially during the busy periods of the day from 10:00 AM until 2:00 PM when multiple users are accessing the Learning Management System (LMS), video conferencing solutions, and cloud services at the same time. The lack of traffic prioritisation makes mission-critical applications like the LMS and financial databases not receive priority over recreational traffic, leading to unacceptable latencies and packet loss for real-time and business-critical applications. Moreover, inadequate network segmentation and security architecture allows for the theft of sensitive administrative and financial information. If the university does not implement a comprehensive redesign of the network, it will continue to suffer from a decline in academic standards, inefficiency, data loss, and reputation.

1.1.2 Background of the University

Greenfield University is a large, multi-campus university with three campuses located in a metropolitan area. Campus A is the main campus with most of the academic departments, administrative offices and central data centre. Campus B and Campus C are smaller satellite campuses, offering extra teaching facilities, research laboratories and student accommodation. The university accommodates around 18,000 students and 2,500 staff (academics, administrative and senior management) spread across 10 main buildings.

The university has a variety of academic programmes in faculties such as Science and Engineering, Business and Law, Health Sciences and Humanities. Research departments that are active in research have applications that are data intensive such as genomic sequencing, climate modelling, and large-scale data visualisation, which demand high bandwidth, low latency connections to centralised computing and storage resources.

Category	Total	Capus A (main)	Campus B	Campus C
Students	18000	10000	5000	3000
Academic Staff	2000	1200	500	300
Administrative Staff	400	250	100	50
Senior Management	100	70	20	10
Total Users	20500	11520	5620	3360

Table 1: University Distribution

1.1.3 Existing Network Environment

The existing network was installed more than 12 years ago and has been incrementally and ad hoc upgraded since then. The architecture is mostly flat and there is little hierarchy between network layers. The current setting has the following features:

- Topology: Single core switch (main campus) as backbone and aggregation point. Campus B and Campus C are single 1 Gbps fibre links to Campus A with no redundancy.
- Switching Infrastructure: Most of the access switches are unmanaged 100 Mbps switches. Distribution-layer switches are a combination of older managed switches that have limited VLAN support and VLAN segmentation is not consistently implemented.
- Routing: Static routing is used in the whole network. No dynamic routing protocol, meaning that if the topology changes, each router must be manually reconfigured.
- Internet Connectivity: 1 ISP internet connection is 2 Gbps and it is supported by a basic stateful firewall, no intrusion prevention or deep packet inspection.
- Mission critical servers, such as the LMS (Canvas), student records database, financial systems, and email servers are all located in one server room at Campus A, known as Data Centre. No DMZ, all servers are on the same flat VLAN as general campus traffic.
- Wireless Coverage: Wireless access points are not centralised and are not always deployed. Wireless access is authenticated using a shared pre-shared key (PSK) instead of enterprise authentication.
- No QOS mechanisms have been implemented. All traffic is treated on a best-effort basis, irrespective of application criticality and/or latency sensitivity.
- Network Management: Only basic reachability checks (ping) are performed. No centralised network management system, no SNMP-based monitoring, no NetFlow analysis, and no automated alerting.

1.1.4 Current Challenges and Limitations

The existing network presents the following critical challenges that the redesign must address:

ID	Challenge	Impact
C1	Insufficient bandwidth between campuses (single 1 Gbps links)	LMS and video conferencing experience severe degradation during peak hours; estimated peak aggregate demand is 10–12 Gbps but only 1 Gbps is available per inter-campus link
C2	No network segmentation or VLAN architecture	All user types (students, staff, guests) share the same broadcast domain; no isolation between administrative, academic, and guest traffic
C3	Absence of QoS and traffic prioritisation	Mission-critical applications (LMS, financial systems) and real-time applications (video conferencing, VoIP) receive the same treatment as recreational streaming and peer-to-peer traffic
C4	Single points of failure throughout the network	A single core switch failure would isolate the entire main campus; a single ISP link failure would cut off all internet access; a single inter-campus link failure would disconnect an entire satellite campus
C5	No DMZ or adequate security architecture	Public-facing servers (university website, LMS gateway, email) are on the internal network with no firewall separation; sensitive financial and student record databases are not isolated from general campus traffic
C6	Static routing with no dynamic failover	Network topology changes require manual reconfiguration; link failures do not trigger automatic rerouting, resulting in prolonged outages
C7	No centralised network monitoring or management	Performance degradation and faults go undetected until users report issues; there is no capacity planning data, no historical traffic analysis, and no SLA measurement
C8	Outdated access layer (100 Mbps unmanaged switches)	Modern applications requiring 1 Gbps or higher to end devices cannot operate effectively; no support for PoE for IP phones or access points
C9	No 802.1X or network access control	Any device can connect to any port without authentication; there is no mechanism to restrict unauthorised or non-compliant devices
C10	No encrypted inter-campus communication	All traffic between campuses traverses the fibre links in cleartext, exposing sensitive administrative and research data to potential eavesdropping

Table 2: Current Challenges and Limitations

1.1.5 Stakeholders

A wide range of stakeholders are involved in the network redesign project, with varied needs and expectations. It is important to recognize and comprehend these stakeholders to ensure the redesigned network is able to satisfy the operational, academic and regulatory requirements of the university.

Stakeholder Group	Representatives	Key Requirements	Primary Concerns
Students	18,000 full-time and part-time students	Reliable access to LMS (Canvas), video conferencing, cloud services, library systems, and internet; minimum 1 Gbps wired connectivity; seamless access across all campuses	Service availability during peak academic periods; latency-sensitive applications (video conferencing) performing adequately; equitable bandwidth allocation
Academic Staff	2,000 lecturers, researchers, tutors	High-speed access to LMS for content delivery; low-latency video conferencing for online teaching; high-bandwidth research data transfers; access to HPC and storage clusters	Research application performance; reliability of teaching tools; consistent access to departmental resources
Administrative Staff	400 finance, HR, registrar, and support personnel	Secure, reliable access to financial systems, student records, payroll, and email; zero tolerance for data loss on financial transactions	Data security and integrity; system uptime during critical processing periods (payroll, billing); compliance with data protection regulations
Senior Management	100 executives, deans, and directors	Visibility into network performance and security posture; strategic reporting and analytics; secure remote access	Risk management; regulatory compliance; cost efficiency; alignment with university strategy
IT Department	Network engineers,	Centralised network management;	Manageability; ability to implement and

	system administrators, security team	automated monitoring and alerting; simplified troubleshooting; scalable architecture that reduces operational overhead	enforce policies; reduction in reactive troubleshooting; support for future growth
External Partners	Research collaborators, cloud service providers	Secure VPN or dedicated connectivity for collaborative research; API access to university systems where appropriate	Data confidentiality; reliable interconnection; compliance with partner security requirements
Students' Family/Guardians	Indirect stakeholders	Access to online fee payment, academic progress portals	Availability of public-facing services; security of financial transactions
Regulatory Bodies	Government education and data protection authorities	Compliance with data protection legislation; audit-ready security controls; incident reporting capabilities	Data breach prevention; privacy of student records; legal and regulatory adherence

Table 3: Stakeholder Analysis

1.1.6 Purpose of the Network Redesign

The aim of this network redesign project is to create a complete, scalable and secure logical network design that meets all the critical issues identified in Section 1.1.4 and satisfies the requirements of all stakeholders identified in Section 1.1.5. In particular, the redesign will seek to:

1. Upgrade the flat and outdated network to a hierarchical core-distribution-access model that offers logical traffic separation, easier troubleshooting and independent scaling of each network layer.
2. Use VLAN segmentation and security zones to separate student, staff, guest, research, and administrative traffic, and to enforce policies with firewall between zones and a DMZ for public facing services.
3. Increase inter-campus and access layer bandwidth to 10 Gbps (inter-campus) and 1 Gbps (access) with redundant links and failover to remove single points of failure and meet the projected peak aggregate demand of 10-12 Gbps.
4. Implement QoS with DiffServ and DSCP marking to prioritise mission-critical and real-time traffic (LMS, video conferencing, financial systems) over best-effort and recreational traffic in congestion.

5. Implement dynamic routing (OSPF) and redundant path diversity for sub-second convergence on link failures and eliminate the operational risk of static route management.
6. Implement centralised network monitoring with SNMP, NetFlow and IP SLA to proactively detect faults, plan capacity and measure SLA compliance.
7. Implement IPsec VPN tunnels for all inter-campus traffic and use 802.1X port-based authentication to provide access control at the device level on all campus networks.

This requirements analysis (Stage 1) forms the basis for the logical network architecture design (Stage 2) and provides a basis for all subsequent design decisions to be traced back to documented requirements, stakeholder needs and identified constraints.

1.2. Project Objectives

1.2.1. Business Goals

The network design for the multi-campus university network is driven by the following business goals:

Goal	Description
1	Support the university's digital transformation strategy by providing reliable, high-speed connectivity across all campuses, enabling modern teaching methods and administrative efficiency.
2	Reduce total cost of ownership by centralizing network management and consolidating services into a primary data centre, eliminating redundant infrastructure across individual campuses.
3	Enable seamless collaboration between campuses, allowing students and staff to access academic resources, attend virtual classes, and communicate with peers regardless of their physical location.
4	Enhance the university's competitive position by offering a technology-rich learning environment that attracts prospective students and retains existing ones.
5	Facilitate hybrid and remote learning models, ensuring that off-campus students have the same quality of access to university resources as those on campus.

Table 4: Business Goals

1.2.2. Technical Goals

The technical objectives of this network design project are:

Goal	Description
1	Design a hierarchical network architecture using a core-distribution-access model to ensure logical separation of traffic and ease of troubleshooting.
2	Implement VLAN segmentation to isolate traffic types including student, staff, guest, research, and administrative traffic across all campuses.
3	Provide redundant links between the core switch and distribution switches at each campus, with automatic failover capability of less than 5 seconds.
4	Establish a centralized data centre that hosts all mission-critical servers, with connectivity to all campuses over redundant 10 Gigabit Ethernet fiber links.
5	Implement a structured IP addressing scheme using VLSM to minimize wasted address space and support route summarization for efficient routing.

Table 5: Technical Goals

1.2.3. Security Goals

The security objectives of this network design are:

Goal	Description
1	Isolate sensitive administrative systems, including financial databases and student records, from general academic traffic using dedicated VLANs and firewall policies.
2	Deploy a next-generation firewall at the internet edge and between the data centre and campus networks to inspect north-south traffic and prevent unauthorized access.
3	Create a DMZ segment for public-facing services such as the university website, LMS, and email gateway to protect internal networks from external threats.
4	Encrypt all inter-campus traffic using IPsec VPN tunnels to prevent eavesdropping and ensure data confidentiality across shared or untrusted WAN links.

5	Implement 802.1X port-based authentication for all wired and wireless access ports to ensure only authorized devices connect to the university network.

Table 6: Security Goals

1.2.4. Scalability Goals

The scalability objectives are:

Goal	Description
1	Support annual user growth of 20 percent for the next three years, reaching approximately 18,000 students and 2,500 staff without requiring a core network redesign.
2	Allow the addition of up to two new campuses over the next five years, with the ability to trunk VLAN's and extend IP addressing without re-addressing existing sites.
3	Design the IP addressing scheme with reserved blocks per campus and per building to accommodate future departmental expansions.
4	Specify modular switch chassis at the core and distribution layers so that line cards and uplinks can be added as bandwidth demands increase.
5	Ensure the routing protocol selection supports the addition of new sites through dynamic route propagation without manual reconfiguration of every router.

Table 7: Scalability Goals

1.2.5. Performance Goals

The performance objectives are:

Goal	Description
1	Deliver end-to-end latency of less than 50 milliseconds between any campus and the central data centre for real-time applications such as video conferencing and VoIP.
2	Provide a minimum of 1 Gigabit per second to every access switch, with 10 Gigabit per second uplinks from each campus distribution switch to the core.

3	Achieve 99.99 percent uptime for mission-critical systems including the LMS, student database, and financial systems, equating to less than 52 minutes of unplanned downtime per year.
4	Maintain less than 1 percent packet loss for high-priority traffic classes during peak utilization hours.
5	Support at least 5,000 concurrent video conferencing sessions across the university with acceptable quality, defined as less than 100 milliseconds latency and less than 30 milliseconds jitter.

Table 8: Performance Goals

2.1.3. Application Requirements

2.1.3.1. Learning Management Systems

The university will deploy a centralized Learning Management System (LMS), such as Blackboard and Canvas to support all teaching and learning activities. The LMS will serve approximately 18,000 students and 2,500 academic staff across three campuses. It hosts course materials, assignments, quizzes, discussions forums, gradebooks, and multimedia content.

Requirement	Value
Estimated bandwidth per concurrent user	Low (0.5 – 1 Mbps), Moderate (1 – 3 Mbps), High (3 – 5+ Mbps) [1]
Maximum latency tolerance	Ideal (< 50 ms), Acceptable (50 – 100 ms), Degrading (> 150 ms) [2]
Jitter tolerance	Ideal (under 20 ms), Acceptable (20 – 50 ms), Threshold (> 50 – 100 ms) [2]
Packet loss tolerance	Ideal (< 0.1%), Acceptable (< 1%), Threshold (> 5%) [1]
Priority level	High
Typical usage pattern	Peak between 9:00 AM – 12:00 PM and 6:00 PM – 9:00 PM

Table 9: Learning Management Systems

1.3 Project Scope

1.3.1 Included Components

This network redesign project will include the following components:

Logical Network Architecture

- Design of hierarchical core-distribution-access network topology for all three campuses (Campus A, Campus B and Campus C)
- VLAN segmentation strategy that separates student, academic staff, administrative, research, guest, and management traffic into separate broadcast domains
- Network addressing architecture (IPv4 subnetting using VLSM and hierarchical addressing), reserved address blocks for future campus and building level expansion
- Redundancy and failover strategies – network routing architecture selection and justification of routing protocols (OSPF for internal routing, BGP for external/internet edge)

Data Centre and Centralised Services

- The central data centre design for Campus A houses all mission critical servers (LMS, student records, financial systems, email and research storage).
- DMZ architecture for public facing services such as the university website, LMS gateway, and email relay
- Segmentation of a server farm using firewall-enforced security zones between internal, DMZ, and external traffic

Security Architecture

- The deployment of next-generation firewall at the internet edge and between the data centre and campus networks.
- All wired and wireless access points are 802.1X port-based authenticated.
- All inter-campus links between Campus A, Campus B, and Campus C are encrypted with IPsec VPN tunnels.
- Access control lists (ACLs) and firewall policies that restrict traffic flow between VLANs and security zones

Network Performance Architecture

- DiffServ implementation with DSCP marking at the access layer, Queuing and Scheduling at the distribution and core layer, Policing at the internet edge and WRED for Congestion Avoidance

- Six traffic classes: Network Control (CS6), Real-Time (EF), Critical Data (AF41), Transactional Data (AF21), Best Effort (BE), and Scavenger (CS1)
- Bandwidth allocation and congestion control strategies for all critical traffic paths including Student VLAN to LMS Server, Staff VLAN to Financial Database, Inter-Campus Backbone, Internet Edge to Cloud Services, Research Lab to High-Performance Storage

Network Management Architecture

- SNMP, NetFlow and IP SLA based centralised monitoring framework for performance measurement, fault detection and capacity planning
- Automated alerting and reporting of all critical network devices and links.
- Monitoring of queue drops, latency per class, and jitter on EF traffic for QoS effectiveness.

Redundancy and Failover

- Redundant 10 Gbps inter-campus links with sub-second convergence, LACP and ECMP load sharing.
- Dual ISP connectivity (ISP 1: primary 5 Gbps, ISP 2: failover) with BGP automatic failover
- Redundant core switches and distribution switches at each campus with automatic failover capability of less than 5 seconds
- Active-passive cluster with stateful failover of less than 3 seconds for financial database path

1.3.2 Excluded Components

This project does not include the following components:

Physical Infrastructure

- Physical cabling installation, fibre trenching and building-to-building cable routing.
- Physical layout of equipment racks, cable management and server room environmental systems (HVAC, power, UPS)
- The cost of specific equipment models, vendor selection and procurement of hardware.

Wireless Network Design

- Wireless access point placement, site surveys and RF planning in detail.
- The following are examples of configuration that is not included in the access layer VLAN and QoS design:

- Wireless mesh or outdoor wireless coverage design.

Application Layer

- Software licenses, deployment or configuration of any software (Canvas LMS, Microsoft Teams, student records system, financial system, etc.)
- Application-layer security (e.g., web application firewalls, database encryption at rest) in addition to network-level security controls

Campus Expansion Detail

- Detailed per room or per floor network design for each building in all three campuses provided, a representative building design provided and scalability demonstrated across additional buildings.
- Network design for any future campus beyond the two planned expansions (the addressing scheme and VLAN design allows for expansion, but specific topology design is left to Stage 2).

Operational Procedures

- Changes in staffing, training or organisational changes
- Beyond network-level redundancy, detailed disaster recovery and business continuity planning.
- Legal, regulatory compliance audits and certification processes

Voice and Telephony

- Only IP telephony QoS requirements are covered, detailed VoIP system design including PBX integration, dial plans and voice mail systems are not covered.

1.3.3 Constraints

The following constraints bound the network redesign project:

ID	Constraint	Description	Impact on Design
CON1	Budget limitation	The university has allocated a fixed capital budget for the network redesign. Cost-effective solutions must be prioritised where performance requirements allow; for example, modular switch chassis at the core and distribution layers to allow incremental upgrades rather than full replacement	Equipment selection must balance cost against the 99.99% uptime requirement; redundant components are specified only where failure would affect mission-critical services
CON2	Existing infrastructure reuse	Approximately 30% of the existing distribution-layer switches and all fibre optic cabling between campuses can be reused, provided they support 10 Gbps uplinks and VLAN trunking	Redesign must account for mixed equipment capabilities during migration; the architecture cannot assume all new hardware at every layer
CON3	Timeline	The project must deliver the logical network design by Week 5 (Stage 1) and the complete architecture by Week 11 (Stage 2), with phased implementation planned across semester breaks to minimise	Design decisions must prioritise solutions that can be implemented in phases; big-bang migration strategies are excluded

		disruption to academic activities	
CON4	Regulatory compliance	The university must comply with national data protection legislation regarding the handling and storage of student records, financial data, and personal information	Security architecture must include encryption at rest and in transit, access control logging, and audit-capable network monitoring; all design choices must support compliance requirements
CON5	Skill availability	The university IT team has limited experience with advanced routing protocols (OSPF, BGP), QoS configuration, and next-generation firewall management	Design must include adequate documentation, and architecture choices should favour industry-standard technologies with well-documented best practices (Cisco and Huawei enterprise campus design guides) over niche or proprietary solutions
CON6	Single data centre	All centralised services are hosted at a single data centre on Campus A; a secondary data centre or disaster recovery site is not within the current project scope	The design must maximise resilience within a single data centre through redundant switches, firewall clusters, and server-level redundancy, while noting that a secondary site is a recommended future enhancement
CON7	Service provider availability	Only two ISPs service the metropolitan area	Internet edge design must work within available ISP

		where the campuses are located, offering combined maximum bandwidth of approximately 5 Gbps (ISP 1: 5 Gbps, ISP 2: lower capacity)	capacity and utilise QoS policing to prevent recreational traffic from consuming disproportionate bandwidth

Table 10: Project Constraints

1.3.4 Assumptions

The requirements analysis and design decisions are based on the following assumptions. All of these assumptions are made when real-world data is not available, and are all supported by reasonable industry benchmarks and the university's context.

ID	Assumption	Justification
A1	The university supports approximately 20,500 total users (18,000 students and 2,400 staff) across three campuses, with annual growth of approximately 20% for the next three years, reaching approximately 24,600 users	Based on university enrolment projections and the scalability goals outlined in Section 1.2.4; the addressing scheme and VLAN design are sized to accommodate this growth without re-architecture
A2	Campus A, Campus B, and Campus C are connected by existing underground fibre optic cabling capable of supporting 10 Gbps links, and a second diverse fibre path (overhead) is available for redundancy	Stated in the inter-campus backbone critical path (Section 2.3.8); assumed based on typical metropolitan university infrastructure where diverse fibre paths exist between campuses
A3	The central data centre at Campus A has sufficient power, cooling, and physical space to accommodate the required server and network equipment upgrades	A physical assessment is outside the scope of this project; the assumption is based on the fact that the existing data centre has been operational for 12 years and typical university data centres have expansion capacity
A4	All end-user devices (workstations, laptops, IP phones) will support 802.1X authentication and DHCP	Modern operating systems (Windows 10+, macOS, major Linux distributions) natively support 802.1X; this is a reasonable assumption given the planned hardware refresh at the access layer
A5	User traffic patterns follow typical university schedules, with peak load between 10:00 AM and 2:00 PM and secondary	Based on traffic load analysis in Section 2.3.6 and typical university network usage studies

	peaks between 6:00 PM and 9:00 PM for LMS access	
A6	The two available ISPs can provide the combined internet bandwidth of approximately 5 Gbps as specified, with BGP support for multihoming	ISP 1 offers 5 Gbps (primary) and ISP 2 offers a backup connection as described in the internet edge critical path (Section 2.3.8); BGP support is standard for enterprise-grade ISP connections
A7	Research data transfers (genomic sequencing, climate modelling) can be scheduled during off-peak hours (11:00 PM to 7:00 AM) to reduce daytime network contention	Research applications typically run long-duration batch jobs; scheduling transfers overnight aligns with the traffic load analysis in Section 2.3.6 and reduces peak-hour bandwidth demand by 1–2 Gbps
A8	The existing fibre optic cabling between campuses is in serviceable condition and does not require replacement, only upgrade of terminations and transceivers to support 10 Gbps	Consistent with the constraint that existing infrastructure will be partially reused (CON2); an assumption that avoids the significant cost and disruption of new fibre installation
A9	Cloud-hosted services (Office 365, Google Workspace, AWS) are accessed via the public internet through the university's internet edge, not via dedicated private peering links	This is the standard access model for most universities; direct peering would require significant additional cost and is excluded from scope (Section 1.3.2)
A10	The university will retain a small number of legacy devices that cannot support 802.1X; these will be placed on a restricted VLAN with limited network access	Practical assumption given the 12-year-old network; legacy devices may include older lab equipment, printers, and building management systems that cannot be immediately replaced

Table 11: Assumptions

1.3.5 Risk Considerations

The following risks have been identified for the network redesign project with mitigation strategies to minimise the likelihood and impact of these risks.

ID	Risk	Likelihood	Impact	Mitigation Strategy
R1	Scope creep leading to project delays and budget overruns	Medium	High	Clearly define included and excluded components (Sections 1.3.1 and 1.3.2); obtain stakeholder sign-off on scope before detailed design begins; enforce change control process for any additions
R2	Budget overruns due to underestimated equipment costs or unforeseen infrastructure requirements	Medium	High	Specify modular equipment that allows incremental upgrades (CON1); phase the implementation across semester breaks to spread costs; include a 15% contingency buffer in budget planning
R3	Single points of failure during migration from the old network to the new architecture	High	High	Implement the new network in parallel with the existing network where physically possible; migrate campus by campus starting with the least critical campus (Campus C); schedule all cutovers during semester break maintenance windows with documented rollback procedures
R4	Data loss or service interruption during cutover from static routing to OSPF	Medium	High	Deploy OSPF in passive monitoring mode before activating it; conduct extensive lab testing of the routing configuration; implement cutover in phased stages (inter-campus links first, then campus-level redistribution) with

				immediate rollback capability
R5	Inadequate IT staff skills for managing new technologies (OSPF, BGP, QoS, next-generation firewalls)	Medium	Medium	Include vendor-provided training in the project budget (CON5); document all configurations with clear operational runbooks; favour industry-standard configurations (Cisco and Huawei best practices) that are well-supported by vendor documentation and community resources
R6	Security breach during the transition period when both old and new network elements are operational	Low	High	Maintain all existing security controls during migration; implement new security zones and firewall policies before decommissioning old controls; conduct penetration testing after each migration phase before opening services to production traffic
R7	QoS misconfiguration causing traffic starvation or priority inversion	Medium	High	Test all QoS policies in a lab environment before production deployment; rate-limit the EF queue to 30% of link capacity to prevent real-time traffic from starving other classes; monitor queue drops, latency per class, and jitter for EF traffic using NetFlow and SNMP; conduct monthly QoS audits
R8	Vendor lock-in limiting future flexibility and cost negotiation	Low	Medium	Design the logical architecture using open standards and vendor-neutral protocols (OSPF, BGP, DiffServ, 802.1X, IPsec); avoid proprietary features where open-standard alternatives exist; ensure all

				configuration templates are portable across major enterprise network vendors
R9	User resistance to new authentication requirements (802.1X) causing support overload	Medium	Medium	Conduct a phased 802.1X rollout starting with staff VLANs, then student VLANs; provide clear documentation, self-service guides, and a dedicated support desk during the transition period; maintain a restricted VLAN for legacy devices that cannot support 802.1X (A10)
R10	Insufficient ISP bandwidth to meet growing external traffic demand	Medium	Medium	Implement QoS policing at the internet edge to guarantee minimum bandwidth for business-critical cloud applications (2 Gbps guaranteed); continuously monitor internet link utilisation using NetFlow; negotiate bandwidth upgrade clauses with ISP contracts in advance of projected demand growth

Table 12: Risk Considerations

2.1 Requirements Specifications

2.1.1 Requirement Gathering Methodology

The requirements for the Greenfield University network redesign were gathered using a multi-method analysis approach to ensure that business, technical, operational, and security requirements were comprehensively identified and validated.

The university operates across three campuses with approximately 20,500 total users and a large number of academic, administrative, research, and cloud-based applications. Due to the scale and complexity of the environment, multiple requirement gathering techniques were used to obtain both qualitative and quantitative data. [3]

Method	Purpose	Description	Stakeholders Engaged	Key Outcomes
Stakeholder Interviews	Identify operational and strategic requirements	One-on-one sessions with department heads	IT Director, Security Manager, Finance Director, Library Dean	Critical business and security requirements identified
User Surveys	Understand user experience and application usage patterns	Distributed questionnaires	Students (sample of 500), Academic staff (200), Administrative staff (150)	Peak usage periods and performance expectations identified
Existing Network Audit	Evaluate limitations of current infrastructure	Analysis of current infrastructure performance logs	IT Operations Team	Bottlenecks, single points of failure, and legacy infrastructure identified
Application Inventory	Identify critical applications and dependencies	Cataloguing all critical applications and their dependencies	Application Owners and department representatives	Mission-critical application mapping completed
Site Walkthroughs	Assess physical distribution of users and services	Physical inspection of buildings across three campuses	Facilities Management	Building-level distribution requirements documented
Traffic Analysis Review	Estimate expected traffic loads and flow patterns	Monitoring and inspecting data packets	Network Administrators	Peak-hour bandwidth and QoS requirements estimated

Table 13: Requirement Gathering Methodology

Requirement Validation Process

All collected requirements were validated against:

- Stakeholder priorities
- Existing infrastructure constraints
- Scalability objectives
- Security requirements
- Expected traffic flows
- Future campus expansion goals

Requirements were then categorised into:

- User Requirements
- Device Requirements
- Network Requirements
- Application Dependencies
- Security Requirements
- Scalability Requirements

Each requirement was assigned:

- A unique requirement ID
- A priority level
- A source
- A campus/building location
- A dependency relationship

This structured methodology ensures that all later design decisions in Stage 2 can be directly traced back to validated operational requirements.

2.1.2 Requirements Template

This Requirements template summarizes the primary application traffic flows within the university network, including user groups, communication protocols, bandwidth consumption, and QoS requirements. This table serves as a high-level reference for the detailed traffic flow analysis presented in the following sections.

Name of Application	**Type of Flow**	**Protocols Used by Application**	**User/User Groups**	**Data Stores (Servers, Hosts, etc.)**	**Bandwidth Requirements**	**QoS Requirements**
Learning Management System (Blackboard)	Internal (Campus -> Data Centre)	HTTPS, TCP/IP	Students, Academic Staff	LMS Server Cluster	0.5-5 Mbps per concurrent user; 4-6 Gbps peak aggregate	High Priority, Low Latency (<100 ms), AF41
Video Conferencing (Collaborate Ultra)	Internal & External	RTP, SRTP, UDP, HTTPS	Students, Academic Staff, Management	Cloud Service Providers	1-8 Mbps per user; high concurrent usage	High priority, low jitter (<30ms), low packet loss (<1%)
Email & Communication Services	External	SMTP, IMAP, POP3, HTTPS	All Users	Microsoft 365/Google Workspace	<1 Mbps per user	Normal Priority
Financial Payroll Systems	Internal (Admin -> Data Centre)	HTTPS, SQL, TCP/IP	Administrative Staff, Management	Finance Database Server	100 Kbps-1.5 Mbps per user; 500 Mbps-1 Gbps peak	Highest Priority, Zero Packet Loss, AF41

Research Applications	Internal (Research Labs -> Storage)	TCP/IP, NFS, SMB	Researchers, Academic Staff	HPC Cluster, Storage Servers	1-2 Gbps peak transfers	Medium Priority, AF21
Library Systems	Internal & External	HTTPS, TCP/IP	Students, Academic Staff	Library Database	~5 Mbps per user	Normal Priority, Low Latency
CCTV & Security Monitoring	Internal	RTP, RTSP, TCP/IP	Security Staff	Library Database Servers	1-20 Mbps per camera	High Priority, Continuous Availablity
Cloud Services	External	HTTPS, TLS	All Users	Cloud Providers	5-25 Mbps per user	Normal Priority
Remote Access VPN	External	IPSec, SSL/TLS	Staff, Students, Researchers,	VPN Gateway	Variable	High Priority, Secure Connectivity
Guest Internet Access	External	HTTPS, DNS	Visitors, External Researchers	Internet Services	Variable	Best Effort

Table 14: Reference for Traffic Flow Analysis

2.1.3 User Requirements

The university network must support approximately 20,500 users distributed across three campuses. User requirements vary depending on operational roles, application usage patterns, security requirements, and mobility expectations.

Req ID	User Group	Count	Campus Distribution	Primary Locations	Key Needs	Expected Concurrent Usage	Priority
UR-01	Students	18,000	Campus A: 10,000, B: 5,000, C: 3,000	Lecture halls, Library, Student Centre, Labs	High-speed Wi-Fi, LMS access, email, cloud storage, video conferencing	70% during peak (12,600 users)	High
UR-02	Academic Staff	2,000	Campus A: 1,200, B: 500, C: 300	Faculty offices, Research labs, Lecture halls	Reliable video conferencing, research data transfer, LMS content creation, remote access	60% during teaching hours (1,200 users)	High
UR-03	Administrative Staff	400	Campus A: 250, B: 100, C: 50	Admin offices, Registrar, Finance, HR	Access to financial systems, payroll, email, document management, VPN for remote work	80% during business hours (320 users)	High
UR-04	Senior Management	100	Campus A: 70, B: 20, C: 10	Executive offices, Admin Building	Secure remote access, dashboards, email, video conferencing	90% during business hours (90 users)	Highest
UR-05	External Researchers/ Visitors	As needed	Distributed across all campuses	Research labs, meeting rooms	Guest Wi-Fi access, limited intranet access, VPN for research collaborators	Variable	Low

Table 15: User Requirements

The user distribution and campus allocation shown above is inspired by the requirements specification format recommended by Tian and Gao [4, p. 74]

User Device Assumptions

Based on current university BYOD trends, each user is expected to connect an average of 1.5 active devices simultaneously, including:

- Laptops
- Smartphones
- Tablets
- Personal IoT devices

This results in an estimated 30,000–35,000 connected devices during peak periods.

User Access Requirements

- The redesigned network must support:
- Seamless roaming across campuses
- Secure remote access via VPN
- Segmented access control using VLANs
- High-density wireless connectivity in lecture halls and libraries
- Role-based access policies for students, staff, and guests

Peak Usage Characteristics

The highest network utilization is expected during:

- 10:00 AM – 2:00 PM (peak academic period)
- 6:00 PM – 9:00 PM (remote LMS and cloud usage)

During these periods, the network must maintain acceptable latency, throughput, and application responsiveness.

2.1.4 Video Conferencing Systems

The university will use video conferencing platforms such as Microsoft Teams, Google Meet, or Zoom for online classes, faculty meetings, guest lectures, and administrative coordination. Video Conferencing is a real-time application that is highly sensitive to network delays and jitter.

Requirement	Value
Estimated bandwidth per concurrent user	Audio only (0.5 Mbps), Standard Definition 480p (1 – 1.5 Mbps), High Definition 720p (2 – 3 Mbps), Full HD 1080p (4 – 6 Mbps), 4K/Multi-stream (> 8 Mbps) [5]
Maximum latency tolerance	Optimal (< 100 ms), Acceptable (< 150 ms), Disruptive (> 200 – 300 ms) [6]
Jitter tolerance	Acceptable (< 30 ms), Maximum (30 – 50 ms) [6]
Packet loss tolerance	Ideal (< 0.5%), acceptable (< 1%), Poor experience (> 3%) [5]
Priority level	High
Typical usage pattern	Scheduled classes, ad-hoc meetings

Table 16: Video Conference Requirements

2.1.5 Email and Communication Services

The university will provide email services (e.g., Google Workspace or Outlook) and instant messaging platforms for all staff and students. This includes calendar services, file attachments, and chat functionality.

Requirement	Value
Estimated bandwidth per concurrent user	< 1 Mbps
Maximum latency tolerance	Seconds to Minutes
Jitter tolerance	Not sensitive
Packet loss tolerance	Moderate
Priority level	Normal

Typical usage pattern	Continuous throughout the day

Table 17: Communication Requirements

2.1.6 Research Applications

Research departments across the university run-data intensive applications including genomic sequencing analysis climate modelling, image processing and large-scale data visualization. These applications often transfer very large files between research labs and central storage or high-performance computing clusters.

Requirement	Value
Estimated bandwidth per concurrent user	80 KBps speed [7]
Maximum latency tolerance	< 50 – 100 ms [7]
Jitter tolerance	Not sensitive
Packet loss tolerance	Moderate
Priority level	Normal
Typical usage pattern	Variable, often scheduled overnight

Table 18: Research Application Requirements

2.1.7 Library Systems

The normal library provides online catalogue access, digital archives, e-journals, research databases, and inter-library loan systems. These services are accessed by students and staff from anywhere on campus or remotely via the internet.

Requirement	Value
Estimated bandwidth per concurrent user	Adequate (5 Mbps), Aspirational (10 Mbps) [7]
Maximum latency tolerance	General application (< 100 ms), interactive application (< 50 ms) [7]
Jitter tolerance	Not sensitive
Packet loss tolerance	Low
Priority level	Normal
Typical usage pattern	Steady throughout the day, peak during exam periods

Table 19: Library Systems Requirements

2.1.8 Financial and Payroll Systems

The university's financial system includes student billing, tuition payment processing, staff payroll, accounts payable and receivable, procurement, and budget management. These systems handle sensitive financial data and transactions.

Requirement	Value
Estimated bandwidth per concurrent user	Recommended allocation 100 Kbps – 1.5 Mbps [8]
Maximum latency tolerance	Optimal performance (< 50 ms), Acceptable performance (< 100 ms), noticeable sluggishness (100 ms), unacceptable (< 250 ms) [9]
Jitter tolerance	Not sensitive
Packet loss tolerance	Zero tolerance
Priority level	Highest (critical)
Typical usage pattern	Periodic (payroll runs, month-end, semester billing)

Table 20: Financial System Requirements

2.1.9 CCTV and Security Systems

The university operates IP-based CCTV cameras across all campuses for security monitoring, along with electronic access control systems for buildings and restricted areas. Video feeds are transmitted to a central security canter for real-time monitoring and recording.

Requirement	Value
Estimated bandwidth per concurrent user	4K (8 – 20 Mbps), 1080p (2 – 5 Mbps), 720p (1 – 2 Mbps) [10]
Maximum latency tolerance	< 500 ms [11]
Jitter tolerance	Moderate
Packet loss tolerance	Moderate
Priority level	Moderate to High

Typical usage pattern	Continuous, 24/7

Table 21: CCTV Requirements

2.1.10 Cloud Services

The university uses various cloud services including Office 365 or Google Workspace for productivity, cloud storage (e.g., OneDrive, Google Drive) and potentially infrastructure-as-a-service (IaaS) or software-as-a-service (SaaS) applications from providers such as AWS, Azure, or specialized educational technology vendors.

Requirement	Value
Estimated bandwidth per concurrent user	5 – 25 Mbps [12]
Maximum latency tolerance	Low (< 20 ms), Moderate (< 100 ms), High (< 200 ms) [12]
Jitter tolerance	Moderate
Packet loss tolerance	Moderate
Priority level	Normal
Typical usage pattern	Distributed throughout the day

Table 22: Cloud Requirements

2.1.11 Device Requirements

The redesigned university network must support a large number of wired; wireless, server, and security devices distributed across all campuses.

The device architecture must support:

1. scalability
2. redundancy
3. centralized management
4. high availability
5. future expansion

Req ID	Device Type	Estimated Quantity	Primary Locations	Network Requirement	Priority
DR-01	Student Workstations	4,000	Computer labs and libraries	1 Gbps wired Ethernet	High
DR-02	Academic Workstations	2,000	Faculty offices and labs	1 Gbps wired Ethernet	High
DR-03	Administrative Workstations	400	Finance, HR, registrar offices	Secure VLAN access	High
DR-04	Student BYOD Devices	20,000+	All campuses	High-density Wi-Fi 6 coverage	High
DR-05	IP Phones	600	Administrative and academic buildings	PoE support and QoS	Medium
DR-06	CCTV Cameras	350	Entrances, corridors, parking areas	Continuous video streaming	High
DR-07	Wireless Access Points	450	Lecture halls, libraries, common areas	Centralized wireless management	High

Table 23: End-User Devices

Req ID	Device Type	Estimated Quantity	Requirement
DR-08	Core Switches	6	Redundant modular chassis with 10–40 Gbps uplinks
DR-09	Distribution Switches	24	Layer 3 switching with VLAN and QoS support
DR-10	Access Switches	120	1 Gbps access ports with PoE support
DR-11	Edge Routers	6	OSPF/BGP support with dual ISP connectivity
DR-12	Next-Generation Firewalls	4	Stateful inspection, IPS, VPN support
DR-13	Wireless LAN Controllers	2	Centralized AP management and authentication

Table 24: Network Infrastructure Devices

Req ID	Device Type	Requirement
DR-14	LMS Server Cluster	High availability with load balancing
DR-15	Financial Database Servers	Redundant storage and restricted access
DR-16	HPC Research Cluster	High-bandwidth internal connectivity
DR-17	Backup Servers	Scheduled backup and disaster recovery support
DR-18	Monitoring Servers	SNMP, NetFlow, Syslog, IP SLA monitoring

Table 25: Data Centre Devices

2.1.12 Requirement Prioritization

Requirements were prioritized according to:

- Business criticality
- Security impact
- Operational dependency
- Performance sensitivity
- Scalability requirements

The prioritization model ensures that mission-critical services receive higher design priority during congestion, failure scenarios, and future expansion planning.

Priority Level	Description	Examples
Highest	Essential for university operations and security	Financial systems, routing protocols, authentication systems
High	Critical academic and operational services	LMS, video conferencing, student records
Medium	Important but can tolerate moderate delay	Email, cloud storage, library systems
Low	Non-critical services	Guest internet, recreational traffic
Lowest	Delay-tolerant background traffic	Backups, bulk updates, P2P traffic

Table 26: Requirement Priority Classification

Requirement ID	Requirement	Priority	Justification
NR-01	Redundant inter-campus backbone	Highest	Failure disconnects entire campuses
NR-02	VLAN segmentation	Highest	Required for security and isolation
NR-03	QoS implementation	High	Protects latency-sensitive traffic
NR-04	802.1X authentication	High	Prevents unauthorized access
NR-05	Centralized monitoring	Medium	Improves operational efficiency
NR-06	Guest internet access	Low	Non-critical service
NR-07	Overnight backup transfers	Lowest	Delay-tolerant traffic

Table 27: Priority Mapping Table

Requirement prioritisation also considered several technical trade-offs:

Trade-Off	Design Decision
Cost vs Performance	Redundancy implemented only for mission-critical systems
Security vs Accessibility	Guest traffic isolated from internal resources
Scalability vs Budget	Modular hardware selected for future expansion
Redundancy vs Complexity	Automatic failover used only on critical paths

Table 28: Requirement Trade-Off Considerations

2.2 Requirements Map

2.2.1 Campus Overview

Greenfield University operates across three campuses connected through a centralized architecture.

- Campus A functions as the primary campus and hosts the central data centre.
- Campus B and Campus C operate as satellite campuses connected through redundant 10 Gbps inter-campus backbone links.

Campus	Type	Students	Academic Staff	Admin Staff	Senior Management	Total Users	Key Buildings
Campus A	Main Campus	10,000	1,200	250	70	11,520	Academic Buildings (multiple), Admin Building, Library, Student Centre, Data Centre, Security Centre
Campus B	Satellite	5,000	500	100	20	5,620	Teaching facilities, Research labs, Student accommodation
Campus C	Satellite	3,000	300	50	10	3,360	Teaching facilities, Research labs, Student accommodation
Total		18,000	2,000	400	100	20,500	10 main buildings total

Table 29: Campus Overview

Building Distribution per Campus (Assumed based on typical university layout):

Building Type	Campus A	Campus B	Campus C
Academic/Lecture Buildings	3	2	2
Research Labs	2	1	1
Administrative Building	1	0	0
Library	1	1	0
Student Centre	1	0	0
Data Centre	1	0	0
Security Centre	1	0	0
Student Accommodation	0	1	1
Total Buildings	10	5	4

Table 30: Building Distribution

The three-campus architecture is inspired by the hierarchical network model described by Tian and Gao [4, pp. 124-128], with Campus A functioning as the core layer and Campuses B and C as distribution/access layers.

2.2.2 Main Campus (Campus A) Application Map

The following table maps each application to the locations where it is consumed (used) and hosted (server location)

Application	Consumer Locations	Dependent User Groups	Host Location
LMS (Canvas)	All campuses, all buildings	Students, academic staff	Campus A Data Centre
Video Conferencing	All campuses, all buildings	Students, Academic Staff, Administrative Staff, External Collaborators	Cloud (Microsoft/Google/Zoom)
Email (Office 365)	All campuses, all buildings	All users	Cloud
Research Applications	Campus A (labs), B (labs), C (labs)	Researchers, Academic Staff, Postgraduate Students, Research Assistants	Campus A Data Centre (HPC)
Library Systems	All libraries, remote access	Students, Academic Staff, Researchers, Library Staff	Campus A Data Centre
Financial & Payroll	Admin Building (Campus A), remote	Administrative staff	Campus A Data Centre (secure)
CCTV & Security	All buildings across 3 campuses	Security staff	Campus A Security Centre
Cloud Services	All campuses, all buildings	All users	Cloud
Student Records	Admin Building, Registrar	Administration, students	Campus A Data Centre
University Website	External internet users	Prospective Students, Current Students, Staff, External Visitors	DMZ (Campus A)

Table 31: Application Map

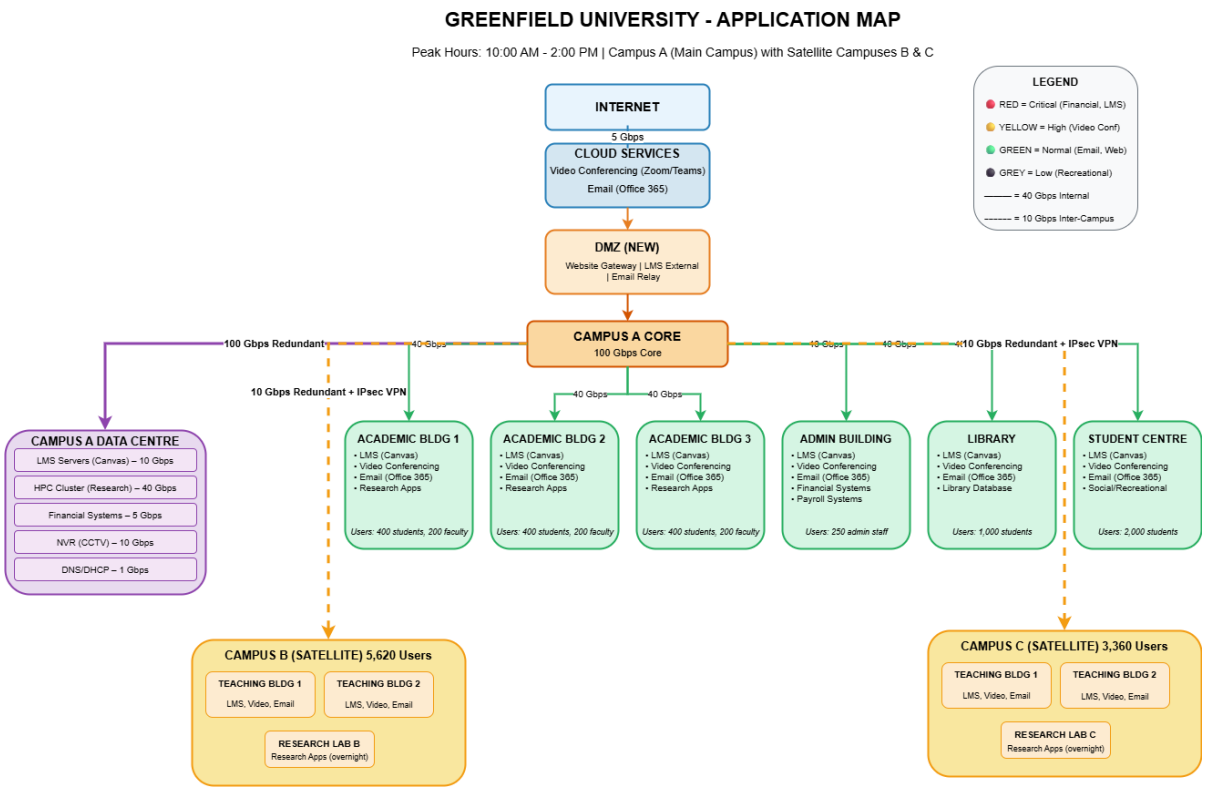


Figure 1: Application Map

The device estimates above follow the methodology described by McCabe [3] for characterising existing network devices and planning for future growth.

2.2.3 Inter-Campus Requirement Dependencies

current inter-campus links are only 1 Gbps (Challenge C1) with no redundancy (Challenge C4), causing severe degradation during peak hours. The redesign must address this:

Traffic Flow	Source Campus	Destination	Current Bandwidth	Required Bandwidth (Projected)	Current Issue	Solution
Campus A ↔ B	Campus B	Campus A	1 Gbps (single link)	10 Gbps	Peak demand exceeds 1 Gbps	Upgrade to 10 Gbps + redundant diverse path
Campus A ↔ C	Campus C	Campus A	1 Gbps (single link)	10 Gbps	Peak demand exceeds 1 Gbps	Upgrade to 10 Gbps + redundant diverse path
Campus B ↔ C	Campus C	Campus B	None (via A only)	10 Gbps (via A)	No direct link; traffic tromboning	Traffic routed via A; future direct link considered
LMS Traffic (B→A)	Campus B	Data Centre (A)	Contends with all traffic	6-8 Gbps during peak	LMS degradation, video freezing	QoS + bandwidth upgrade
Research Traffic (B→A)	Campus B	HPC (A)	Overnight only	2-3 Gbps	Daytime transfers impossible	Schedule overnight + dedicate bandwidth

Table 32: Inter-Campus Requirement Dependencies

Dependent Campus	Dependency	Impact if Link Fails
Campus B	LMS access	Teaching disruption
Campus B	Financial systems	Administrative disruption
Campus C	Student records	Enrollment disruption
Campus C	Cloud access	Collaboration disruption

Table 33: Critical Dependencies

Design Implications

These dependencies require:

1. Redundant backbone links
2. Dynamic routing
3. QoS prioritization
4. Failover protection
5. Continuous monitoring

2.2.4 Building-Level Requirement Distribution Diagram

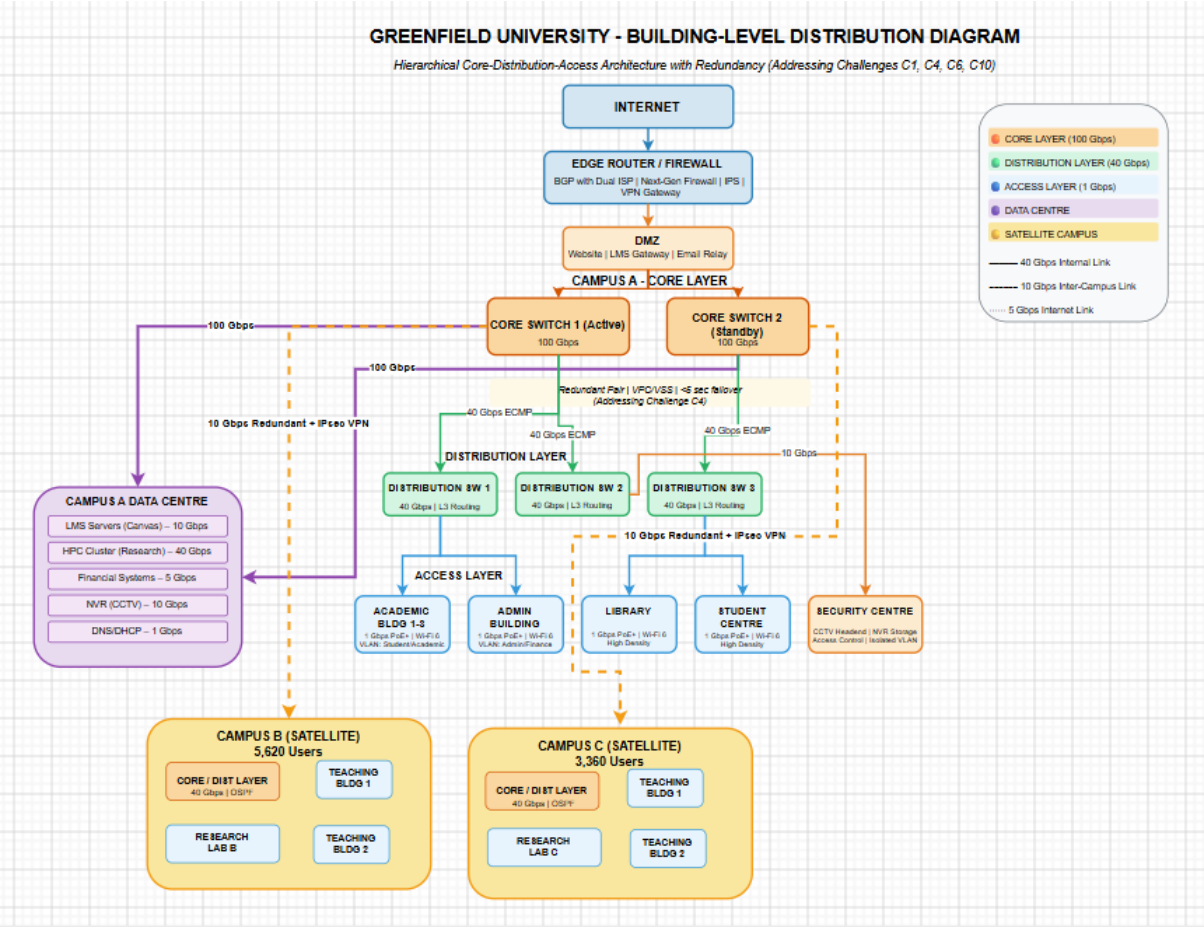


Figure 2: Building-Level Distribution Diagram

Architecture Notes:

- Hierarchical core-distribution-access model across all three campuses
- Redundant inter-campus links at 10 Gbps (addressing Challenge C1 and C4)
- IPsec VPN encryption for inter-campus traffic (addressing Challenge C10)
- Redundant core switches at Campus A with <5 second failover
- OSPF dynamic routing for sub-second convergence (addressing Challenge C6)

2.2.5 Data Centre (Campus A) Requirement Mapping

Application	Server Location	Bandwidth Need	Redundancy Required	Security Zone (New Design)	Current Issue	Solution
LMS (Canvas)	Data Centre Rack A	10 Gbps	Active-Passive	Internal VLAN + DMZ gateway	On flat VLAN with no isolation	Move to Internal VLAN, DMZ gateway for external access
Student Records	Data Centre Rack A	5 Gbps	Active-Passive	Restricted VLAN	On flat VLAN, unauthorised access risk	Isolated VLAN with strict ACLs
Financial/Payroll	Data Centre Rack C (secure cabinet)	5 Gbps	Active-Active	Finance VLAN (isolated, encrypted)	On flat VLAN, data theft risk (Challenge C5)	Dedicated secure zone, zero-tolerance for packet loss
Research HPC	Data Centre Rack B	40 Gbps	None (checkpoint/backup)	Research VLAN	Contends with general traffic	Dedicated VLAN, QoS prioritisation
Library Systems	Data Centre Rack A	5 Gbps	Backup nightly	Internal VLAN	No isolation	Move to Internal VLAN
Security NVR	Data Centre Rack D	10 Gbps	RAID storage	Security VLAN	On flat VLAN	Isolated VLAN, continuous 24/7 traffic
DNS/DHCP	Data Centre Rack A (virtual)	1 Gbps	Redundant VMs	Management VLAN	No dedicated management network	Out-of-band management
DMZ Services	DMZ Rack	5 Gbps	Active-Passive	DMZ (isolated from internal)	No DMZ exists (Challenge C5)	New DMZ for website, LMS gateway, email relay

Table 34: Requirement Mapping

2.2.6 Scalability Across All Three Campuses

The design scales across Greenfield University's three campuses as follows:

Campus	Type	User Count	Connection to Main Campus	Application Access	Design Approach
Campus A	Main Campus	11,520	N/A (core)	Full access to all on-prem and cloud apps	Full three-tier design with redundant core, multiple distribution switches
Campus B	Satellite	5,620	10 Gbps redundant links + IPsec VPN	Full access to on-prem apps via WAN; local caching for LMS	Distribution-core model with redundant links to Campus A
Campus C	Satellite	3,360	10 Gbps redundant links + IPsec VPN	Full access to on-prem apps via WAN; local caching for LMS	Distribution-core model with redundant links to Campus A

Table 35: Scalability Approch

Application	Location Dependency	Scaling Implication
LMS (Canvas)	Low latency to Campus A Data Centre	Campus B and C need 10 Gbps links + local caching servers to reduce WAN load
Video Conferencing	Cloud-based, no location dependency	Works anywhere with sufficient internet bandwidth (5 Gbps primary)
Financial Systems	Must remain in Campus A Data Centre	Campus B/C access requires secure WAN + IPsec VPN
Research Applications	Large data transfers to/from Campus A HPC	Schedule overnight transfers (11pm-7am) to reduce daytime contention
CCTV	Local recording at each campus? (Assumption)	Each campus needs local NVR; central viewing via WAN
Email/Cloud	Internet-dependent	Each campus needs reliable internet failover

Table 36: Location Dependency Considerations

Requirement	Scalability Goal
VLAN architecture	Extend to future campuses
IP addressing	Reserved subnets for expansion
Routing protocols	Dynamic route propagation
Backbone design	Additional campuses supported
Wireless architecture	Centralized expansion capability

Table 37: Scalability Requirements

Expansion Strategy

The architecture reserves:

- additional VLAN ranges
- additional subnet blocks
- scalable routing areas
- modular switch capacity

This enables the university to add up to two additional campuses without major restructuring.

2.2.7 Requirement Dependency Analysis

According to the identified challenges in section 1.1.4, the following dependencies must be addressed:

Requirement	Depends On	Impact if Dependency Fails
LMS access (Student)	Core switch (Campus A), Inter-campus links (10 Gbps), Data Centre	No classes, high impact; 10-12 Gbps peak demand unmet
Video Conferencing	Internet uplink (5 Gbps primary), QoS	Classes disrupted, but recordable
Financial transactions	Secure VLAN, Data Centre, Firewall, Redundancy	Payroll/billing halted, critical; data breach risk
CCTV recording	PoE switches, Security Centre, Bandwidth	Security blind spots; continuous 24/7 requirement
Inter-campus traffic	10 Gbps links, OSPF routing, IPsec VPN	Campus B/C isolated; unencrypted data exposure
Remote access (VPN)	Edge firewall, Internet, ISP availability	Staff cannot work from home
Network monitoring	SNMP, NetFlow, management VLAN	Faults undetected; no capacity planning
802.1X authentication	RADIUS server, device compatibility	Unauthorised device access; security risk

Table 38: Requirement Dependency Analysis

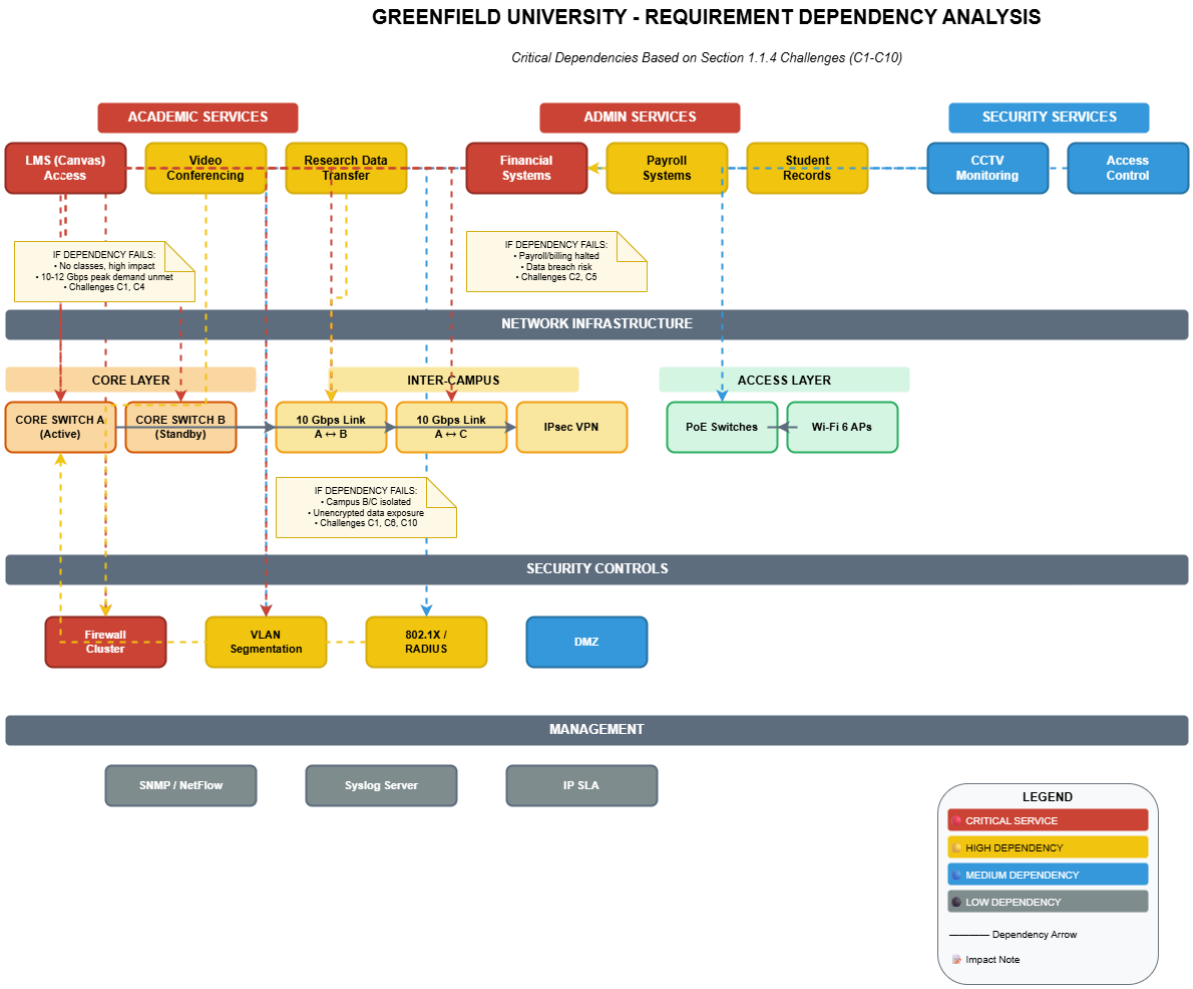


Figure 3: Requirement Dependency Analysis

[4]

2.3 Traffic Flow Specifications

2.3.1 Traffic Flow Overview

The multi-campus university network carries five major categories of traffic:

Category	Description
Academic Traffic	LMS, library systems, research data, transfers, and educational content delivery.
Administrative Traffic	Financial systems, payroll, student records, human resources systems, and email communications.
Real-time Traffic	Video conferencing for online classes and meetings, VoIP call, and live streaming of lectures.
Security and Surveillance Traffic	CCTV video streams, access control system, communications, and security alert data.
Guest and Recreational Traffic	Visitor internet access, student recreational streaming, social media, and personal device usage.

Table 39: Traffic Flow Categories

Traffic flows can be categorized as internal (campus-to-campus, building-to-building, or VLAN-to-VLAN) and external (campus-to-internet and campus-to-cloud services). Most mission-critical traffic flows between the campuses and the central data centre, where core servers are hosted.

2.3.2 Internal Traffic Flows

Internal traffic flows consist of communication between devices within the university network that does not exit to the internet. The primary internal flows include:

Flow Direction	Description	Typical Volume
Student VLAN to LMS Server	Students accessing course materials, submitting assignments, and taking quizzes	High during peak academic hours
Staff VLAN to Finance DB	Administrative staff querying and updating financial records	Moderate, periodic
Research Lab to Storage Server	Large file transfers for research projects	Variable, often high
Classroom to Media Server	Lecture capture uploads and streaming	Moderate during class times
Building to Building	General traffic between different buildings on the same campus	Moderate
Campus to Data Centre	All traffic destined for centralized servers	Very high

Table 40: Primary Internal Flows

Internal traffic is generally preferred over external internet paths, meaning internal routes should be prioritized and kept as short as possible for latency-sensitive applications.

2.3.3 Non-Real-Time Traffic

Non-real-time traffic can tolerate variable delays and does not have strict delivery deadlines. This traffic type is typically handled using best-effort queuing. In the university network, non-real-time traffic includes:

1. Email transmission and retrieval
2. Web browsing (non-interactive)
3. File downloads (software updates, drivers)
4. Backup data transfers
5. Print jobs
6. Bulk research data transfers (when not time-sensitive)

Non-real-time traffic does not require special QoS treatment beyond normal best-effort delivery. However, during congestion, non-real-time flows, such as overnight backups can be scheduled during off-peak hours to reduce daytime network contention.

2.3.4 Traffic Load Analysis

Based on a typical multi-campus university with 18,000 students and 2,500 staff, the expected traffic load is estimated as follows:

Time Period	Activity	Expected Load
8:00 AM – 10:00 AM	Morning classes begin, email check, LMS access	Medium to High
10:00 AM – 02:00 PM	Peak class period, video conferencing, simultaneous LMS access	Highest
02:00 PM – 05:00 PM	Afternoon classes, research activities, administrative work	Medium to High
05:00 PM – 08:00 PM	Evening classes, recreational internet use	Medium
08:00 PM – 11:00 PM	Peak recreational traffic (streaming, gaming, social media)	High
11:00 PM – 07:00 AM	Backups, updates, maintenance	Low

Table 41: Peak Hours Identification

Traffic Category	Estimated Peak Aggregate	Direction
LMS and academic	4-6 Gbps	Campus to Data Centre
Video conferencing	2-3 Gbps	Campus to Data Centre and Internet
Administrative systems	500 Mbps – 1 Gbps	Campus to data Centre
Research transfers	1-2 Gbps	Lab to Data Centre
Guest and recreational	2-4 Gbps	Campus to Internet
Cloud services	1-2 Gbps	Campus to Internet

Table 42: Aggregate Bandwidth Estimate

The core network should be provisioned to handle approximately 10-12 Gbps of aggregate traffic during peak periods, assuming standard oversubscription ratios.

2.3.5 QoS Requirements

The university network will implement QoS (Quality of Service) using Differentiated Services (DiffServ) with DSCP marking at the access layer. QoS is essential to ensure that latency-sensitive and mission-critical traffic receives preferential treatment during congestion, while best-effort traffic yields appropriately [13].

2.3.5.1. QoS Traffic Classes and Marking:

Traffic Class	DSCP Marking	Queue Type	Bandwidth allocation	Use Case
Network Control	CS6 (48)	Strict Priority	5%	OSPF, BGP, STP, routing protocols
Real-Time (Voice/Video)	EF (46)	Low Latency Queue (LLQ)	30%	Video conferencing, VoIP, live streaming
Critical Data	AF41 (34)	Weighted Fair Queue (WFQ)	35%	LMS interactive, financial database, student records
Transactional Data	AF21 (18)	Weighted Fair Queue	15%	Email, library catalogue, web browsing
Best Effort	BE (0)	Default FIFO Queue	10%	Guest internet, recreational traffic
Scavenger	CS1 (8)	Default FIFO Queue	5%	Backups, P2P file sharing, software updates

Table 43: QoS Traffic Classes and Marking

2.3.5.2. QoS Implementation Strategy:

1. Classification and Marketing (Access Layer):

All classification and DSCP marking will be performed at the access switches as traffic enters the network. Access control lists (ACLs) and Network-Based Application Recognition (NBAR) will identify application traffic based on IP addresses, protocols, and port numbers. Trust boundaries will be enforced at access ports, and any existing DSCP markings from end-user devices will be overwritten to prevent abuse.

2. Queuing and Scheduling (Distribution and Core Layers):

Distribution and core switches will implement LLQ for EF-class traffic, ensuring that real-time packets are transmitted before any other queue. The LLQ is rate-limited to 30% of link capacity to prevent starvation of other traffic classes. Assured Forwarding (AF) queues will use Weighted Fair Queuing (WFQ) with configured bandwidth percentages. Best Effort and Scavenger traffic will share the remaining bandwidth using default FIFO queuing.

3. Policing and Shaping (Network Edge):

Ingress policing will be applied at the internet edge firewall to limit guest VLAN traffic to 2 Gbps, preventing recreational traffic from consuming all internet bandwidth. Egress shaping will be configured on inter-campus 10 Gbps links to smooth bursty research to traffic, reducing the likelihood of tail drops.

4. Congestion Avoidance:

Weighted Random Early Detection (WRED) will be enabled on all AF queues. WRED proactively drops lower-priority packets within each class before the queue becomes full, preventing global synchronization and maintaining throughout for higher-priority flows within the same class.

1. Link Efficiency:
On inter-campus WAN Links, compression and fragmentation mechanics will be considered for VoIP traffic, though the primary focus remains or proper queuing and adequate link capacity.

Verification and Monitoring:

QoS effectiveness will be monitored using NetFlow and SNMP. Key metrics include queue drops, latency per class, and jitter for EF traffic. Regular audits will ensure DSCP markings are preserved end-to-end and no device is remarking or stripping QoS bits.

The QoS design is based on Cisco and Huawei best practices for enterprise campus networks [13][14].

2.3.6 Bottleneck Analysis

Based on the traffic patterns described above, the following potential bottlenecks have been identified:

Bottleneck	Description
Inter-Campus Links	The links connecting branch campuses to the main campus and data centre are the most critical points in the network. If these links become saturated, all traffic to and form remote campuses will suffer. During peak hours, LMS traffic alone consume 40-50 percent of a 10 Gbps link, leaving limited capacity for video conferencing and other flows.
Internet Edge	The university's internet connection is shared by all external traffic including guest access, student recreational traffic, cloud services, and staff web browsing. Without QoS policies, guest and recreational traffic could overwhelm the internet link and degrade critical external services such as cloud-hosted email or video conferencing.
Data Centre Aggregation Switches	All traffic from all campuses converges at the data centre aggregation switches before being directed to individual servers. Oversubscription at this layer can cause tail drops and retransmissions, particularly affecting real-time flows.
Access Layer to Distribution Layer Uplinks	In dense areas such as large lecture halls or library study spaces, many users share a single uplink from the access switch to the distribution layer. During class changes or exam periods, this uplink can become congested, causing localized performance degradation.
Wireless Network Controllers	While this design focuses on wired infrastructure, it is worth noting that wireless access points share backhaul capacity. Many students simultaneously connecting to WiFi in a

	single building can overwhelm the wired uplink long before the wireless spectrum is exhausted.

Table 44: Bottleneck Analysis

2.3.7 Traffic Flow Prioritization Table

The following prioritization scheme will be applied to all traffic flows across the university network:

Priority Level	Queue Class	Traffic Types	Bandwidth	Drop Preference
Level 1 (Highest)	EF (Expedited Forwarding)	Network control protocols (OSPF, STP, BGP), VoIP	5 - 10%	Never drop
Level 2 (High)	AF4 (Assured Forwarding)	Video conferencing, live streaming, LMS interactive	30 – 40%	Drop only when queue full
Level 3 (Medium)	AF2 (Assured Forwarding)	Financial systems, student records, email, library	25 – 30%	Drop before high priority
Level 4 (Low/Normal)	AF1 (Assured Forwarding)	Research file transfers, web browsing, cloud sync	15 – 20 %	Drop before medium priority
Level 5 (Lowest)	BE (Best Effort)	Guest internet, recreational streaming, backups, P2P	Remaining bandwidth	Drop first during congestion

Table 45: Prioritization Scheme

This prioritization ensures that:

1. Network stability traffic (routing protocols) always gets through
2. Real-time teaching and collaboration tools are protected
3. Administrative systems maintain acceptable performance
4. Best-effort traffic yields during congestion

2.3.8 Traffic Flow Diagrams

Diagram 1: Traffic Flow Overview

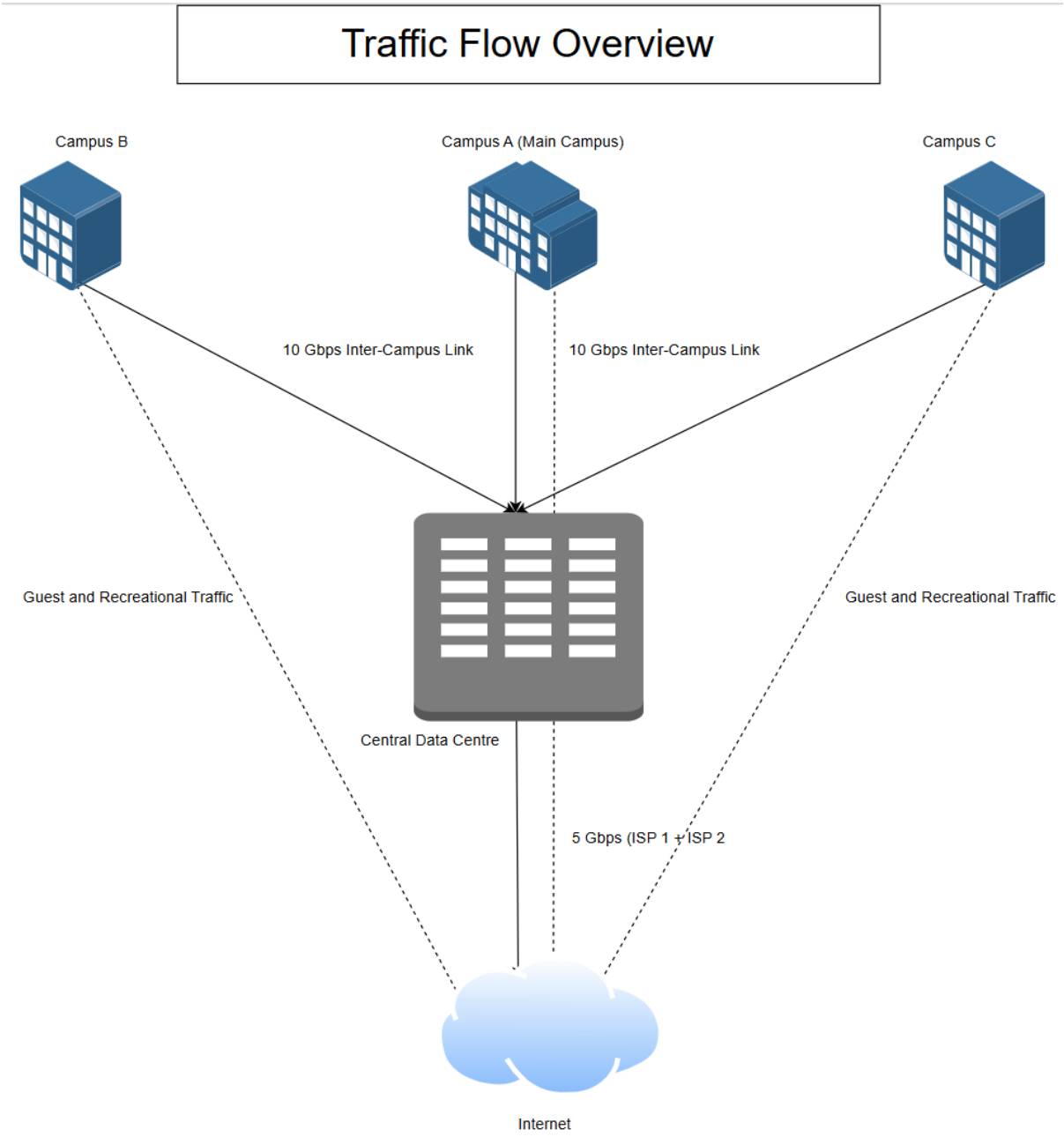


Figure 4. High-level traffic flow overview showing three campuses, central data centre, and internet connectivity.

Explanation: This diagram provides a high-level visualization of the university network's physical topology and major traffic flows. Three campuses are shown: Campus A (main campus), Campus B, Campus C. All three campuses connect to the Central Data Centre via redundant 10 Gbps inter-campus links, which carry most of the academic and administrative traffic including LMS, financial systems, and email. The Data Centre hosts mission-critical servers and connects to the internet through dual 5 Gbps ISP links. Guest and recreational traffic flows directly from campuses to the internet, by

passing the Data Centre where it is possible to reduce load on internal links. Green arrows represent internal traffic, while red arrows represent external-bound traffic. This topology ensures that all campuses have equal access to centralized resources while maintaining clear separation between internal and external traffic paths.

Diagram 2: LMS Traffic Path (End-to-End)

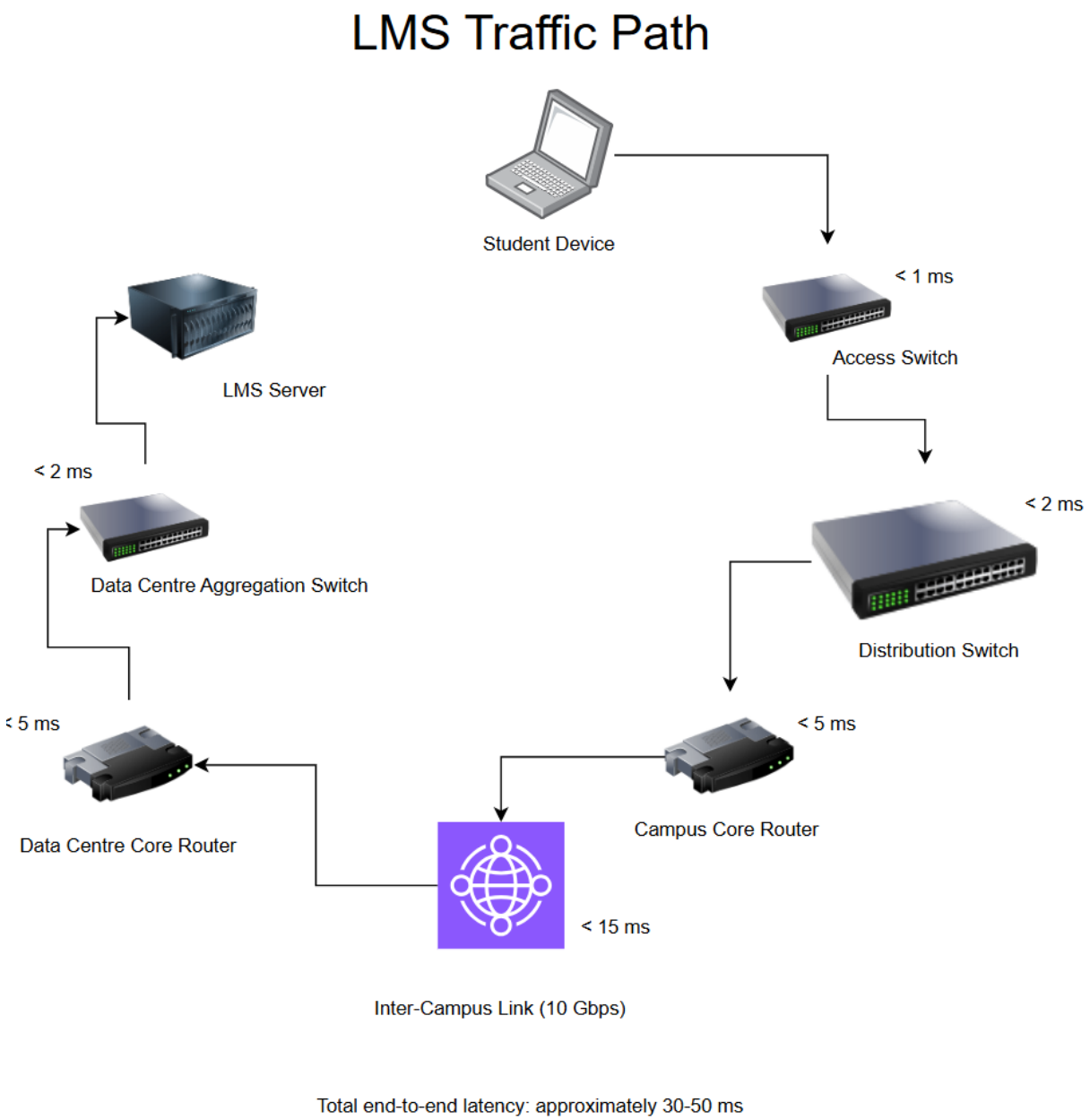


Figure 5. End-to-End LMS Traffic Path from student device to LMS server with per-hop latency estimates.

Explanation: This diagram traces the complete path of a student request to the request to the Learning Management System (LMS) server, from the moment it leaves a student device until it reaches the server in the Data Centre. The path traverses eight network devices:

Student Device > Access Switch > Distribution Switch > Campus Core Router > Inter-Campus WAN Link > Data Centre Core Router > Data Centre Aggregation Switch > LMS Server.

Each hop is labelled with its estimated latency contribution, ranging from less than 1 ms at the access switch to approximately 15 ms across the inter-campus fiber link. The total end-to-end latency is approximately 30 to 50 ms, which satisfies the performance goal of less than 50 ms for real-time applications. This path is critical because LMS access directly impacts the university's academic operations.

Diagram 3: QoS Marking and Queuing Points

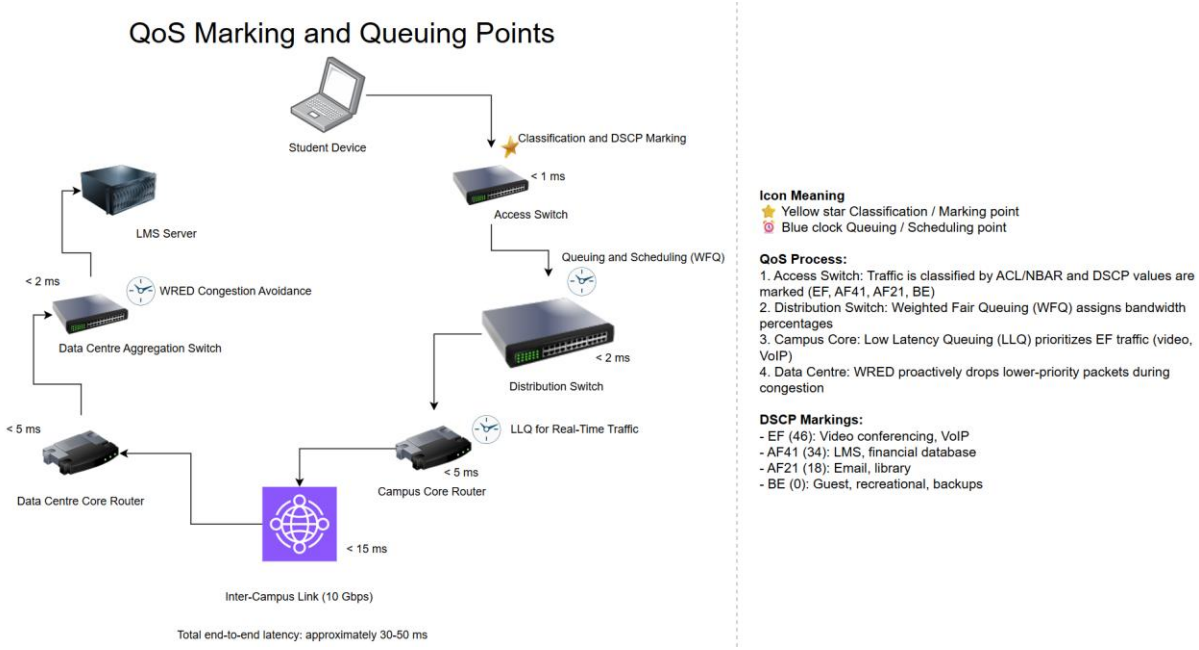


Figure 6. QoS marking and queuing points across the LMS Traffic Path.

Explanation: This diagram builds upon the LMS traffic path shown in Figure 2, adding QoS decision points to illustrate where traffic classification, marking, queuing, and congestion avoidance or NBAR, and DSCP values are assigned (EF for real-time, AF41 for critical data, AF21 for transactional data, and BE for Best Effort). At the Distribution Switch and Campus Core Router (marked with green clocks), Weighted Fair Queuing (WFQ) and Low Latency Queuing (LLQ) are applied, LLQ gives strict priority to EF-marked traffic such as video conferencing. At the Data Centre Aggregation Switch, Weighted Random Early Detection (WRED) proactively drops lower-priority packets before congestion occurs. This end-to-end QoS strategy ensures that latency-sensitive and mission-critical traffic receives preferential treatment, while best-effort traffic yields during congestion.

Diagram 4: Bottleneck Analysis

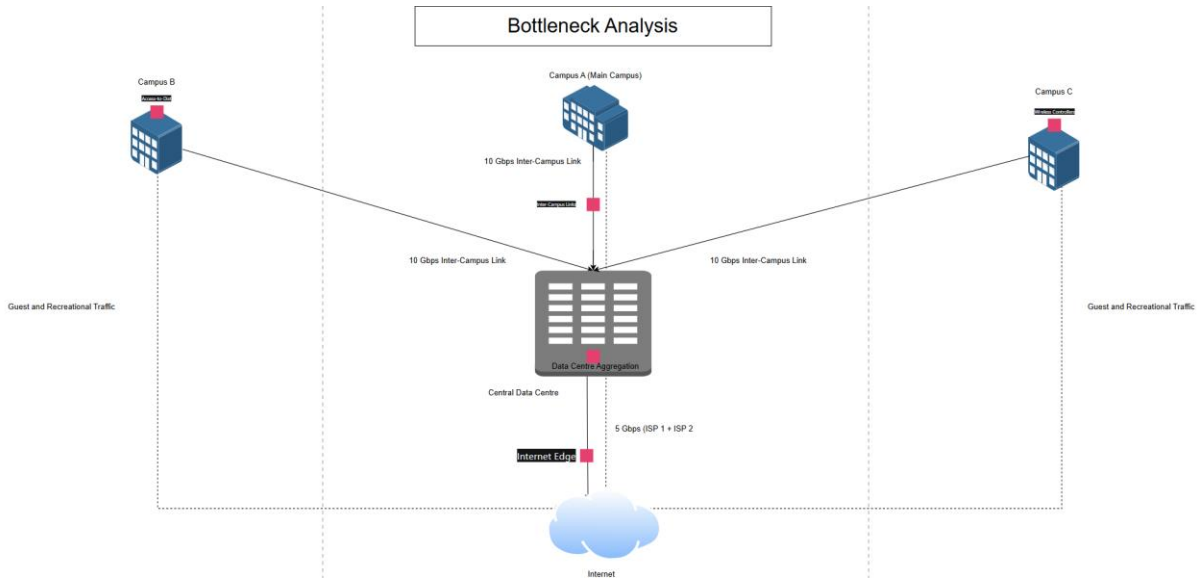


Figure 7. Bottleneck analysis with five critical congestion points marked in red.

Explanation: This diagram highlights the five most likely congestion points in the university network, overlaid on the high-level topology from Figure 1. Each red marker corresponds to a specific bottleneck identified in Section 2.3.9:

Bottleneck 1 (Inter-Campus Links):

LMS traffic alone consumes 40-50 percent of 10 Gbps link capacity during peak hours, leaving limited bandwidth for other flows.

Bottleneck 2 (Internet Edge):

Guest and recreational traffic can overwhelm the 5 Gbps internet connection if not policed, degrading cloud services.

Bottleneck 3 (Data Centre Aggregation):

All traffic from all campuses converges at aggregation switches, where oversubscription can cause tail drops.

Bottleneck 4: (Access-to-Distribution Uplinks):

High-density areas such as lecture halls and libraries have many users sharing a single uplink.

Bottleneck 5: (Wireless Controllers):

Many simultaneous WiFi connections can overwhelm wired backhaul capacity. Each bottleneck is colored red (severity for high), and a summary table lists mitigation strategies such as adding third links, upgrading uplinks, and implementing traffic policing.

Diagram 5: Internet Edge and External Traffic

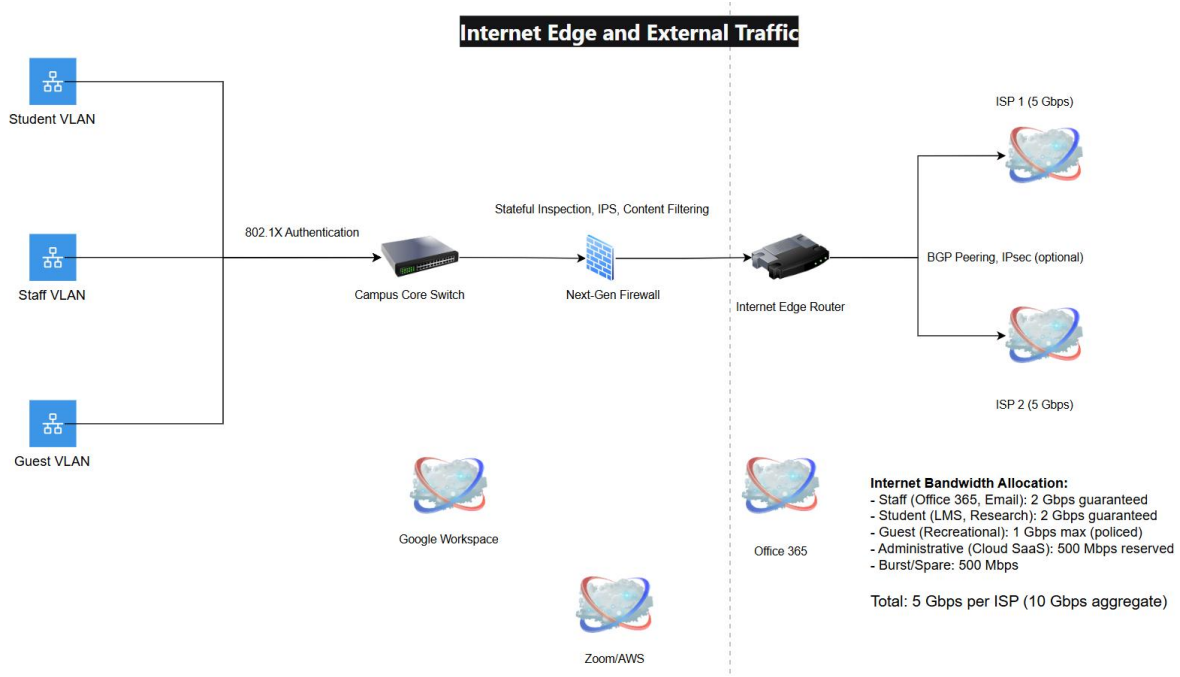


Figure 8. Internet Edge architecture showing VLANs, firewall cluster, dual ISPs, and cloud service connectivity.

Explanation: The diagram illustrates how external-bound traffic flows from internal VLANs to the internet and cloud services. Three VLANs are shown: Student VLAN, Staff VLAN, and Guest VLAN, all aggregating at the Campus Core Switch. Traffic then passes through an active-active firewall cluster providing stateful inspection, intrusion prevention, and content filtering. The Internet Edga Router connects to dual ISP links (ISP 1 and ISP 2, each 5 Gbps) running BGP for automatic failover. From there, traffic reaches cloud services including Office 365, Google Workspace, and Zoom/AWS. Guest traffic is separately policed to a maximum of 2 Gbps to prevent recreational usage from impacting academic traffic. The dashed return arrows indicate inbound traffic following the same reverse path. Bandwidth allocation is summarized in the accompanying text box: Staff traffic receives 2 Gbps guaranteed, Student traffic 2 Gbps guaranteed, Guest traffic receives 2 Gbps, Administrative cloud services 500 Mbps reserved, with 500 Mbps remaining for burst or spare capacity.

2.4 Network Requirements

Based on the identified user and application requirements, the proposed university network must provide a secure, scalable, high-performance, and high available infrastructure capable of supporting approximately 20,500 users across three campuses. The network must efficiently support both real-time and non-real-time services while ensuring reliable connectivity, acceptable Quality of Service, and strong security protections for academic, administrative, and research operations.

2.4.1 Bandwidth Requirements

The university network must provide sufficient bandwidth capacity to support high volumes of concurrent users and bandwidth-intensive applications across all campuses. As identified in Section 2.1.3, applications such as video conferencing, cloud services, and multimedia learning systems generate significant network traffic during peak academic hours.

To accommodate this demand, the network backbone should support high-speed fibre connectivity with a minimum backbone capacity of 10 Gbps between core network devices and inter-campus links. Access layer connections within academic buildings, laboratories and administrative offices should support at least 1 Gbps Ethernet connectivity to prevent local congestion.

Wireless infrastructure must also support high-density user environments such as lecture theatres, libraries, and student centers where thousands of simultaneous wireless connections may occur. The network should therefore support Wi-Fi 6 of newer wireless standards to improve throughput, device density handling, and spectral efficiency.

Additionally, sufficient bandwidth allocation must be reserved for research-related traffic and cloud-based services to prevent performance degradation during periods of high utilization.

2.4.2 Availability and Reliability Requirements

The university network must maintain high availability and reliability to ensure uninterrupted access to critical academic and administrative services. Applications such as the Learning Management Systems (LMS), payroll systems, financial databases, and security monitoring services are considered missions-critical and require continuous operation.

The network should therefore minimize downtime through the implementation of fault-tolerant infrastructure and resilient network design principles. Redundant hardware components, backup communication links, and high-availability configurations must be implemented to reduce the impact of hardware failures or service interruptions.

Furthermore, network monitoring and automated fault detection mechanisms should be used to identify and resolve connectivity issues quickly. Reliable connectivity is especially important during examination periods, online assessments, payroll processing cycles, and scheduled virtual learning sessions.

2.4.3 Redundancy Requirements

To eliminate single points of failure, the university network must incorporate redundancy at both the device and link levels. Core switches, routers, internet connections, and critical servers should operate in redundant configurations to ensure service continuity in the event of equipment failure.

Dual internet service provider connections should be deployed to maintain external connectivity if one provider becomes unavailable. Similarly, inter-campus communications links should include failover paths to preserve connectivity between campuses and the central data centre.

Critical systems such as databases, authentication servers, and LMS platforms should also implement data redundancy and backup replication mechanisms to support disaster recovery and business continuity objectives.

2.4.4 Security Requirements

The network must provide comprehensive security protections to safeguard sensitive academic, financial, and administrative information. As identified in previous sections, systems such as payroll services, student databases, cloud platforms, and remote-access services contain confidential institutional data and must therefore be protected against unauthorized access and cyber threats.

The proposed network should implement layered security controls including enterprise firewalls. Intrusion detection and prevention systems, network access control mechanisms, and secure VPN connectivity for remote users.

Network segmentation using VLANs should also be implemented to isolate sensitive systems and restrict unnecessary traffic between departments. Role-Based Access Control policies should enforce the principle of least privilege by ensuring users can only access resources required for their operational responsibilities.

In addition, Multi-Factor Authentication should be implemented for administrative systems and remote-access systems and remote-access services to strengthen identify verification and reduce the risk of credential compromise.

2.4.5 Scalability Requirements

The university network must support future growth in student enrollment, connected devices, digital services, and campus expansion. The infrastructure should therefore adopt a modular and scalable architecture capable of accommodating increasing traffic demands without major redesign.

Scalability considerations include support for additional wireless access points, future campus buildings, increased cloud service integration, and the growing use of Internet of Things devices such as smart sensors, surveillance systems, and automated classroom technologies.

The network should also support scalable IP addressing schemes and flexible VLAN structures to simplify future expansion and administrative management.

2.4.6 Quality of Service Requirements

The university network must support Quality of Service mechanisms to prioritize latency-sensitive and mission-critical traffic. As identified in Section 2.1.3, real-time applications such as video conferencing and security monitoring systems are highly sensitive to latency, jitter, and packet loss.

Quality of Service policies should therefore prioritize real-time traffic over lower-priority services such as general web browsing and non-urgent file transfers. Traffic classification, bandwidth reservation, and packet prioritization techniques should be implemented to maintain acceptable application performance during periods of heavy network utilization.

Additionally, Quality of Service mechanisms should help ensure stable communication quality for online lectures, virtual meetings, and live security monitoring operations across all campuses.

2.4.7 Wireless Coverage Requirements

Comprehensive wireless coverage must be provided throughout all university campuses to support student mobility, flexible learning environments, and Bring Your Own Device practices. Wireless access should be available in lecture halls, libraries, research laboratories, administrative buildings, outdoor learning spaces, and student common areas.

The wireless network should support seamless roaming between access points to maintain uninterrupted connectivity while users move across campus locations. High-density deployment strategies must also be implemented in areas with large concentrations of concurrent users to reduce congestion and maintain stable wireless performance. Furthermore, wireless security mechanisms such as WPA3-Enterprise

authentication should be implemented to protect user communications and prevent unauthorized network access.

2.4.8 Remote Access Requirements

The university network must support secure remote connectivity for students. Academic staff, researchers, and administrative personnel accessing university resources from off-campus locations. Remote access capabilities are essential for hybrid learning, online collaboration, remote administration, and external research partnerships.

Secure VPN services should be deployed to encrypt remote communications and protect sensitive institutional data transmitted over public networks. Remote users should also be authenticated using Multi-Factor Authentication to strengthen access security.

Additionally, the network should support reliable access to cloud-based services, virtual learning environments, email platforms, and shared resources regardless of user location.

2.5 Technical Trade-Off Analysis

This section evaluates key design trade-offs involved in implementing the proposed university network. These trade-offs arise due to competing requirements such as performance, cost, security, scalability, and operational complexity. The objective is to justify balanced design decisions that meet institutional needs while remaining feasible within real-world constraints.

2.5.1 Cost vs Performance

A major trade-off in the network design is balancing infrastructure cost against performance. High-performance networking components such as enterprise-grade routers, Layer 3 switches, Wi-Fi 6/6E access points, and redundant fibre links significantly improve throughput and reliability but also increase capital and operational expenditure.

To meet the performance expectations of high-bandwidth applications, the university must prioritize investment in core and distribution layer infrastructure while optimizing costs at the access layer through scalable deployment strategies. A hierarchical network design helps balance this trade-off by concentrating high-performance hardware at critical network points while using cost-effective device at edge locations.

2.5.2 Security vs Accessibility

The network must balance strong security controls with user accessibility. Implementing strict authentication mechanisms, network segmentation, and access control policies improves protection of sensitive systems such as financial databased and student records.

However, overly restrictive security measures may reduce usability, especially for students and staff requiring seamless access to learning platforms, cloud service, and remote systems.

To address this trade-off, a layered security approach should be adopted. This includes role-based access control, secure VPN access for remote users, and VLAN segmentation to isolate sensitive traffic without impacting general usability. This ensures security is maintained without significantly hindering legitimate access.

2.5.3 Scalability vs Budget Constraints

Scalability is essential to accommodate future growth in student population, device density, and application demand. However, designing a fully scalable infrastructure from the outset can be costly.

To manage this trade-off, the network design should adopt a modular and incremental scalability approach. This includes using a scalable IP addressing scheme, expandable

switching infrastructure, and cloud-based services that allow capacity expansion without significant hardware investment.

This approach ensures that the network can grow over time while aligning with budget limitations and phased investment strategies.

2.5.4 Redundancy vs Complexity

Introducing redundancy improves network reliability and fault tolerance by ensuring alternate paths and backup systems are available in case of failure. However, excessive redundancy increases network complexity, management overhead, and cost.

For this university network, redundancy should be applied strategically rather than universally. Core systems such as data centres, inter-campus links, and internet gateways should include full redundancy, while less critical edge components may use simplified configurations.

This selective redundancy approach ensures high availability for mission-critical services without unnecessarily increasing operational complexity.

2.5.5 Performance vs Manageability

High-performance network configurations often require advanced routing protocols, load balancing mechanisms, and dynamic traffic management, which can increase administrative complexity.

To balance this trade-off, the network should adopt standardized configuration models and centralized network management tools. Automation and monitoring systems can help reduce administrative burden while maintaining high performance.

This ensures that the network remains effective and responsive without becoming overly complex to manage.

2.5.6 On-Premises Infrastructure vs Cloud Integration

Another key trade-off involves choosing between on-premises infrastructure and cloud-based services. On-premises systems offer greater control, lower latency for internal services, and enhanced data governance. However, they require higher upfront investment and ongoing maintenance.

Cloud services provide scalability, flexibility, and reduced infrastructure maintenance but may introduce dependency on external providers and variable latency.

A hybrid approach is therefore recommended, where critical systems such as authentication and internal databases are hosted on-premises, while scalable applications such as email, collaboration tools, and storage leverage cloud platforms.

3. Preliminary Design Considerations

This section outlines the initial logical design direction for the proposed multi-campus university network. The design is based on the previously identified user requirements, application demands, network requirements, and technical trade-offs. It focuses on establishing a scalable, secure, and high-performance architecture capable of supporting current operations and future expansion.

3.1 Proposed Hierarchical Network Model

The university network should be implemented using a hierarchical three-layer architecture consisting of the Core Layer, Distribution Layer, and Access layer. This model is widely adopted in large-scale enterprise and campus networks due to its scalability, modularity, and ease of management.

Core Layer: Serves as the high-speed backbone of the network and is responsible for interconnecting all campuses and major data centers. It must be optimized for high availability, low latency, and high throughput. Redundant core switches and high-capacity fibre links should be implemented to ensure uninterrupted connectivity between campuses.

Distribution Layer: Aggregates traffic from multiple access switches and applies routing, filtering, and policy enforcement. This layer plays a critical role in implementing inter-VLAN routing, Quality of Service policies, and security controls. It also ensures efficient traffic distribution between different departments and services.

Access Layer: Provides direct connectivity to end-user devices such as laptops, desktops, IP phones, printers, and IoT devices. This layer supports wired and wireless access and should be designed for high device density, particularly in lecture halls, libraries, and student areas.

3.2 Initial Security Architecture Considerations

The security design for the university network must follow a layered security approach to protect sensitive academic, financial, and administrative data.

Internal network segmentation should be achieved using VLANs to separate different user groups such as students, academic staff, administrative staff, and guests. This limits unauthorized lateral movement within the network.

Key security mechanisms include:

- Enterprise firewalls at network boundaries.
- Intrusion Detection and Prevention Systems.
- Multi-Factor Authentication for privileged access.
- Role-Based Access Control for system permissions.

- Secure VPN for remote access users.

This layered approach ensures confidentiality, integrity, and availability of university systems while maintaining usability.

3.3 Future Expansion and Scalability Considerations

The network must be designed with scalability as a core principle to accommodate future growth in users, devices, applications, and campus locations.

Scalability is supported through:

- Modular network design allowing incremental expansion.
- Scalable IP addressing scheme (supporting IPv4 growth and IPv6 readiness).
- Flexible VLAN architecture for adding new departments or buildings.
- Upgradable core infrastructure supporting higher bandwidth demands.
- Support for additional campuses without major redesign.

Cloud integration further enhances scalability by enabling dynamic resource allocation for services such as storage, email, and learning platforms.

This ensures the network can evolve alongside the university's academic and technological growth without significant structural redesign.

3.4 IPv6 Readiness and Addressing Strategy

To support long-term scalability and address exhaustion limitations of IPv4, the network should be designed with IPv6 compatibility.

IPv6 adoption provides:

- A vastly larger address space.
- Improved routing efficiency.
- Better support for mobile and IoT devices.
- Reduced reliance on NAT

A dual-stack approach is recommended during the transition phase, allowing both IPv4 and IPv6 to operate simultaneously. This ensures backward compatibility while preparing the network for future expansion.

A structured hierarchical UP addressing scheme should also be implemented to simplify network management and troubleshooting across campuses.

3.5 Cloud Integration and Hybrid Architecture Considerations

The proposed network should adopt a hybrid architecture combining on-premises infrastructure with cloud-based services.

On-premises infrastructure will support:

- Internal databases (student records, finance systems)
- Authentication services
- Internal file sharing and research systems.

Cloud services will support:

- Email and collaboration platforms.
- Scalable storage solutions
- Learning management systems
- Software-as-a-service applications

This hybrid approach provides a balance between performance, control scalability, and cost efficiency.

Secure connectivity to cloud services should be ensured using encrypted channels, VPN tunnels, and secure API integrations where applicable.

4. Conclusion

The following is a summary of the key operational, technical, security, scalability and performance issues identified in this requirements analysis for the Greenfield University network redesign project. The current network environment is bandwidth constrained, lacks redundancy, has limited security controls, has poor traffic prioritisation and has outdated infrastructure, all of which have a negative effect on teaching, research, administration, and users' experience.

The project analysed the requirements of the stakeholders, traffic flows, application dependencies, device requirements and scalability requirements to provide a solid basis for the future logical network design. The proposed redesign focuses on implementing a hierarchical core-distribution-access architecture, VLAN segmentation, dynamic routing, QoS mechanisms, redundant high-speed inter-campus links, enhanced security controls, and centralized monitoring systems. The enhancements will ensure reliable connectivity, secure access to essential services, fault tolerance, and support future university expansion.

Additionally, the analysis took into account practical project constraints, assumptions, and risks to ensure that the proposed architecture is realistic, scalable, and meets the university's operational and financial needs. Moreover, the prioritisation of mission-

critical applications like the LMS, financial systems, video conferencing, and research applications ensures that the re-designed network can deliver high availability and performance during peak usage.

Overall, this Stage 1 requirements analysis will serve as a comprehensive and traceable basis for Stage 2 of the project, in which the detailed logical network architecture and implementation strategies will be developed. The new network will enhance the university's ability to meet the current and future needs of its users, while also improving the security, resilience, and performance of the network.

5. Individual Reflection

My Contribution

In Assignment 1, I was responsible for defining the project objectives and documenting the application requirements, completing the traffic flow specifications, and creating the supporting diagrams.

Written deliverables I completed

- Five categories of project objectives (Business, Technical, Security, Scalability, and Performance Goals) with measurable targets including 99.99% uptime and sub-50ms latency
- Eight application requirement profiles (LMS, video conferencing, email, research apps, library systems, financial systems, CCTV, cloud services) with bandwidth, latency, jitter, packet loss, and priority specifications.
- Complete traffic flow specifications including internal/external traffic tables, real-time vs non-real-time characterization, peak hour identification (10 AM – 2 PM), aggregate bandwidth estimates, QoS table with DSCP markings, five critical traffic paths with protection mechanisms, and prioritization tables.

Figures/Diagrams I created (using Draw.io)

- Traffic Flow Overview: illustrating five major categories across the multi-campus network.
- LMS Traffic Path: showing the complete path from student access switches to data centre servers with redundancy points.
- QoS Marking and Queuing Points: mapping where classification, marking, and queuing occur across access, distribution, core, and WAN layers.
- Internet Edge and External Traffic: depicting dual ISP connections, firewall cluster, and cloud service destinations.
- Bottleneck Analysis Diagram: visualizing five key congestion points (inter-campus links, internet edge, data centre aggregation, access-distribution uplinks, wireless controllers)

I also designed the cover page for the document.

My Peers' Contribution

Maria established the project introduction, scope, constraints, and assumptions, giving me clear boundaries for my traffic analysis. Saha gathered user and device requirements, built the application map, and handled building-level requirement distributions, her user counts across buildings directly informed my bandwidth calculations. Natalee defined network requirements (bandwidth, availability, redundancy, security) and led technical trade-off analysis, helping me justify my QoS queue allocations.

Challenges Faced and Resolved

The main challenge I faced was ensuring my traffic estimates and QoS allocations aligned with my peers' data. My initial bandwidth calculations did not match Saha's building distribution, and my first QoS table summed to 105% instead of 100%. I resolved this by scheduling team syncs to cross-reference our assumptions, revising my calculations twice, and consulting Cisco and Huawei best practice guides to correct the QoS percentages.

What I Learned

I learned that network diagrams are not decorative, they must be technically accurate and aligned with written specifications. A QoS table without a corresponding diagram showing where marking occurs is incomplete. I also gained practical skills in Draw.io, including using layers for complex diagrams and consistent icon sets for network devices. Finally, I learned that traffic flow specifications require iteration, my first version had to be revised twice to align with Maria's scope boundaries and Saha's building distribution before submission.

6. References

[1]

Huawei.com, 2026. https://support.huawei.com/enterprise/en/doc/DOC1000091883/cb4be28b/introduction (accessed May 10, 2026).

[2]

K. Carlson, “What is Latency?,” Internet Speed Test - Check Your Speed | TestMySpeed.com, Apr. 04, 2023. https://www.testmyspeed.com/insights/what-is-latency

[3]

J. D. McCabe, Network Analysis, Architecture, and Design, 3rd ed. Amsterdam: Morgan Kaufmann Publishers, 2007.

[4]

Y. C. Tian and J. Gao, Network Analysis and Architecture. Singapore: Springer, 2024.

[6]

G. Melville, “What Bandwidth Do Companies and Users Need for Video Conferencing? - Personal SASE - The #1 Corporate VPN Alternative,” Personal SASE - The #1 Corporate VPN Alternative, Apr. 10, 2025. https://cloudbrink.com/blog/bandwidth-for-video-conferencing-2/

[7]

JKaval, “Acceptable Jitter, Latency and Packet Loss for Audio and Video on a WebEx Meeting,” Cisco.com, Mar. 04, 2021. https://community.cisco.com/t5/webex-meetings-and-webex-app/acceptable-jitter-latency-and-packet-loss-for-audio-and-video-on/m-p/4301454 (accessed May 10, 2026).

[8]

R. N. De Silva, W. Cheng, W. T. Ooi, and S. Zhao, “Towards understanding user tolerance to network latency and data rate in remote viewing of progressive meshes,” Proceedings of the 20th international workshop on Network and operating systems support for digital audio and video, pp. 123–128, Jun. 2010, doi: https://doi.org/10.1145/1806565.1806595.

[9]

Oqitor, “Oqitor | Enterprise Technology Procurement Partner | 200+ Suppliers,” Oqitor, 2026. https://www.oqitor.com/resources/bandwidth-requirements (accessed May 10, 2026).

[10]

“Zero-Tolerance Latency: Why Financial Services Demand Integrated Edge Security and Acceleration - Tencent Cloud,” Tencentcloud.com, 2026. https://www.tencentcloud.com/techpedia/143884?lang=en (accessed May 10, 2026).

[11]

“Verkada - Enterprise Security Camera System - No NVR or DVR Required,” www.verkada.com. https://www.verkada.com/

[9]

“Strategies for Fostering Collaboration and Alignment with VSM,” Broadcom.com, 2024, doi: https://doi.org/1078097655/broadcom-academy-unified4.

[12]

L. Morley, “High-Bandwidth Use Cases Now Cost-Effective on Private Cloud,” OpenMetal IaaS, Jan. 27, 2026. https://openmetal.io/resources/blog/high-bandwidth-use-cases-now-cost-effective-on-private-cloud/

[13]

M. ALshawi, “Campus Network Design Guideline,” community.cisco.com, Oct. 14, 2012. https://community.cisco.com/t5/networking-knowledge-base/campus-network-design-guideline/ta-p/3140160

[14]

Huawei.com, 2026. https://support.huawei.com/enterprise/en/doc/EDOC1000069520/d1ec1156/campus-networks-typical-configuration-examples (accessed May 10, 2026).

Link to the Video

https://www.youtube.com/watch?v=NR8P7zbERGI